

A Review on Intelligent Scheduling and Optimization for Flexible Job Shop

Bin Jiang* , Yajie Ma, Lijun Chen, Binda Huang, Yuying Huang, and Li Guan

Abstract: Flexible job shop scheduling problem is a NP-hard combinatorial optimization problem, which has significant applications in the field of workshop scheduling and intelligent manufacturing. Due to its complexity and significance, lots of attention have been paid to tackle this problem. This paper reviews some of the researches on this problem, by presenting and classifying the different criteria, constraints, and solution approaches. The existing solution methods for the flexible job shop scheduling problem in this literature are classified into exact algorithms, heuristics, and meta-heuristics, which are thoroughly reviewed. Particularly, the paper highlights the flexible job shop scheduling problem in the context of dynamic events and preventive maintenance. These dynamic events, such as machine breakdowns and unexpected changes in job requirements, present additional challenges to the scheduling problem. Furthermore, this paper analyzes the development trends in the manufacturing industry and summarizes detailed future research opportunities for the flexible job shop scheduling problem.

Keywords: Dynamic scheduling, flexible job shop scheduling, intelligent optimization, preventive maintenance, static scheduling.

1. INTRODUCTION

Manufacturing industry is one of the largest industrial sectors, providing a large number of employment opportunities for the society [1]. Scheduling is the core issue in manufacturing industry and has been widely studied. The purpose of scheduling is to consider various constraints in production to allocate available production resources to jobs, so that the optimization objective is optimal [2], and the most considered optimization objective is processing time. Job shop scheduling problem (JSP) is one of the famous production scheduling problems, which is a NP-hard problem [3]. In JSP, a set of jobs is processed on a limited set of machines and each operation is processed on a machine with a given processing time [4].

As an extension of JSP, flexible job shop scheduling problem (FJSP) is more realistic, but also more complex. FJSP can be described as: Completing the processing task of n jobs on m machines, each job consists of no less than one operation. Any operation can be processed on one or more machines, but the processing efficiency of different machines is different. The goal of this problem is to arrange a reasonable processing sequence for the

operations and allocate the most suitable processing machine to make the optimization objective optimal under various constraints. Therefore, FJSP mainly includes two sub-problems: machine selection and process sequencing. This problem is called FJSP in Brandimarte [5], job shop scheduling problem for multi-purpose machines in Hurink, Jurisch and Thole [6], multiprocessor job shop scheduling problem in Dauzère-Pérés & Paulli [7], and parallel machine job shop scheduling problem in Liu & Kozan [8]. The basic problem description of FJSP will be given in Section 2.

Small-scale FJSP can find the optimal scheduling by enumeration method, while the complexity of large-scale FJSP increases sharply with the increase of the number of machines and jobs. Therefore, finding an effective scheduling method has become a problem that researchers need to overcome. In general, the scheduling problem can be divided into three aspects: exact algorithm, heuristics and meta-heuristics. Static scheduling methods are summarised and categorised in Section 3.

Static scheduling, although widely used, fails to account for the dynamic elements inherent in the production process. Unforeseen disturbances during manufactur-

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ing can disrupt the schedule, leading to interruptions or even full stops in production. To address these challenges, dynamic scheduling emerges as a crucial process for updating scheduling solutions in response to dynamic events that occur in the production process. Therefore, dealing with FJSP under dynamic disturbances is of great significance and we provide an overview of it in Section 4.

Machine breakdowns are the most talked about disruption in FJSP. Reactive scheduling is real-time and efficient which rearranges available resources after a machine failure, but it ignores the high reliability requirements of manufacturing systems, such as aerospace industry system. In addition, reactive scheduling reallocates multiple resources, which may lead to system chaos. More importantly, reactive scheduling does not fundamentally reduce the failure rate of the system. Preventative maintenance regularly overhauls machines and repairs them before they breakdown, which can radically improve the robustness of the system, a comprehensive summary is given on preventive maintenance for FJSP in Section 5.

Based on our review, some interesting summaries and perspectives for future research are given in Section 6.

2. PROBLEM DESCRIPTION

The problem of FJSP is described as follows: a batch of jobs $J = \{J_1, J_2, \dots, J_n\}$ is to be processed on a set of m machines $M = \{M_1, M_2, \dots, M_m\}$; each job consists of one or more operations; the sequence of operations is pre-determined; each operation can be processed on multiple different machines; the processing time of each operation varies depending on the machine; the scheduling objective is to select the most suitable machine for each operation, determine the optimal processing sequence and start time for each operation on each machine, and optimize certain performance indicators of the entire system. Therefore, the flexible job shop scheduling problem consists of two sub-problems: determining the machines for each job (machine selection sub-problem) and determining the processing sequence on each machine (operation sequencing sub-problem). Additionally, the following constraints need to be satisfied during the processing:

The mathematical model of the flexible job shop scheduling problem is defined as follows: n : total number of jobs. m : total number of machines. Ω : total set of machines. i, e : machine indices, $i, e = 1, 2, 3, \dots, m$. j, k : job indices, $j, k = 1, 2, 3, \dots, n$. h_j : total number of operations for job j . l : machine indices, $i, e = 1, 2, 3, \dots, h_j$. Ω_{jh} : set of available machines for the h th operation of job j . m_{jh} : number of available machines for the h th operation of job j . O_{jh} : h th operation of job j . M_{ijh} : processing of the h th operation of job j on machine i . p_{ijh} : processing time of the h th operation of job j on machine i . s_{jh} : start time of the h th operation of job j . c_{jh} : completion time of the h th operation of the j th job. If operation O_{ijh} is assigned to

machine i , then it is 1; otherwise, it is 0. If operation O_{ijh} is processed before operation O_{ikl} , then it is 1; otherwise, it is 0.

$$s_{ij} + x_{ij} \times p_{ij} < c_{jh}, \quad (1)$$

$$c_{jh} < s_{j(h+1)}, \quad (2)$$

$$c_{jh_j} < C_{\max}, \quad (3)$$

$$s_{jh} + p_{ijh} \leq s_{kl} + L(1 - y_{ijhkl}), \quad (4)$$

$$c_{jh} \leq s_{j(h+1)} + L(1 - y_{ikl j(h+1)}), \quad (5)$$

$$\sum_{i=1}^{m_{jh}} x_{ijh} = 1, \quad (6)$$

$$\sum_{j=1}^n \sum_{h=1}^{h_j} y_{ijhkl} = x_{ikl}, \quad (7)$$

$$\sum_{k=1}^n \sum_{l=1}^{h_k} y_{ijhkl} = x_{ijh}. \quad (8)$$

Equations (1) and (2) indicate the constraints before and after each process, and (3) for the job completion time constraints, the completion time of each job can not exceed the total completion job time. Equations (4) and (5) means that the same machine can only process one process, (6) means that the same process can only be processed by one machine, (7) and (8) means that each machine is operating in a cycle.

The aim of the scheduling solution is to achieve the optimum of the corresponding objective. Common optimisation objectives in the context of workshop scheduling and intelligent manufacturing include maximum completion time, maximum machine load, total machine load, energy consumption, and lead time etc.

Min makespan: Completion time refers to the time required to finish a job, while makespan represents the time taken to machine the longest job, serving as a fundamental performance indicator in research on FJSP. It is defined as follows:

$$\min(\max(C_i)). \quad (9)$$

Min maximum machine load: Total machine load refers to the sum of time taken by all machines to carry out the processes. A higher machine load indicates that a machine takes a longer time to complete a processing task. It is defined as follows:

$$\min \left(\sum_{i=1}^n \sum_{j=1}^{n_i} \sum_{k=1}^m t_{ijk} \times x_{ijk} \right). \quad (10)$$

Min total machine load: The machine with the highest load becomes the bottleneck, and it is essential to balance the load on each machine participating in the process to improve performance. It is defined as follows:

$$\min \left(\max \sum_{i=1}^n \sum_{j=1}^{n_i} t_{ijk} \times x_{ijk} \right). \quad (11)$$

3. SOLUTION METHODS FOR STATIC FLEXIBLE JOB SHOP SCHEDULING PROBLEM

The design of solutions for the FJSP is much more complex compared to the JSP. In the FJSP, it is necessary to consider not only arranging appropriate sequence for each operation, but also arranging appropriate machines for each operation, which makes the calculations more complicated. Especially when the problem scale expands, the existing algorithms inevitably face issues of low computational efficiency and poor solution quality.

The classical FJSP is an academic optimization problem in the field of workshop scheduling. In order to effectively model and solve practical scheduling problems, it is necessary to extend the FJSP to incorporate additional constraints and characteristics. Such as dual-resource constrained FJSP (DRCFJSP), and distributed FJSP. Solving the classical FJSP considering the constraints mentioned in Section 2 is extremely difficult. Building on this, it is a great challenge to solve the DRCFJSP and the distributed FJSP considering factory distribution. This section provides a comprehensive summary of the classical FJSP, DRCFJSP, and distributed FJSP.

3.1. Classical flexible job shop scheduling problem

The design of FJSP solutions becomes much more complex compared to JSP due to the flexibility of machine allocation. This section provides a comprehensive summary of the classical FJSP, which includes single objective classical FJSP and multi-objective classical FJSP.

3.1.1 Approaches for single objective classical flexible job shop scheduling problem

The methods used to solve FJSP can be categorized into three types: exact algorithms, heuristics, and meta-heuristic methods. Fig. 1 depicts the distribution of the three approaches used for the single objective FJSP. Meta-heuristic algorithm takes up the largest proportion, accounting for approximately half of the total. It is followed by the heuristic algorithm and the exact algorithm. Compared to exact algorithms, heuristic algorithms can solve FJSP more quickly. When it comes to tackling the most difficult instances of FJSP, advanced meta-heuristics are found to be the most suitable approaches.

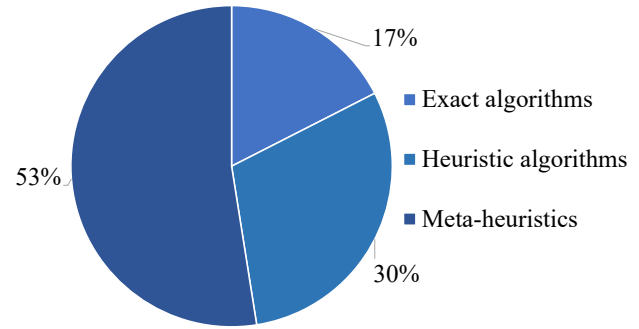


Fig. 1. Distribution of approaches for single objective FJSP.

The objective function and approach of single FJSP is shown in Table 1. The most frequent objective is the makespan, occurring 35 times (89.7%) [9-52]. The makespan is expressed by (9). Apart from the makespan, other commonly seen criteria in scheduling literature have been relatively underexplored. Such as average tardiness [17], total cost [20], and total weighted tardiness [18]

$$\text{Average tardiness} = \frac{\sum_{i=1}^n \max(0, C_i - d_i)}{n}, \quad (12)$$

$$\text{Total cost} = \sum_{i=1}^n c(f_i), \quad (13)$$

$$F = \sum w_i T_i, \quad (14)$$

where C_i is the completion time of the job i , and d_i is the due date of job i , w_i is the weight of the job i , T_i is the tardiness of job i , $c(f_i)$ is the cost of job i .

Exact algorithms: Previous research has presented exact algorithms for solving FJSP. Many studies on FJSP focus on developing integer linear programming (ILP) or mixed integer linear programming (MILP) models [9-12]. In [9], Özgüven *et al.* developed a mixed-integer linear programming model (MILP-1) and compared it to an alternative model from the existing literature to demonstrate its superior computational efficiency. In [10], Elazeem *et al.* introduced theorems and lemmas to define optimality conditions for FJSP solutions and presented a mathematical model for FJSP. In [11], Roshanaei *et al.* developed two novel effective position and sequence based MILP

Table 1. Objective function and approach of single objective FJSP.

Approach	Objective function	Number	Reference
Exact algorithms	Makespan	7	[9-15]
Heuristic algorithms	Makespan	8	[16,19,21-25,27]
	Total cost	2	[20,26]
	Average tardiness	1	[17]
	Total weighted tardiness	1	[18]
Meta-heuristics	Makespan	20	[29-37,39,40,42,44,45,47-52]

models. An auto MILP (Auto-MILP) model for FJSP was proposed in [12].

Branch and bound (B&B) algorithms are the main exact algorithms considered for solving FJSP optimally, both through dedicated algorithms and by modeling the problem as a MILP [13–15]. In [13], Zhou *et al.* constructed Petri net models for FMS and obtained optimal schedules using the B&B algorithm. In [14], Lloyd *et al.* proposed an optimum scheduling algorithm for FMS using Petri net modeling and modified B&B search. In [15], Hansmann *et al.* derived a mixed-integer linear optimization model and proposed a B&B procedure. The solution obtained by the exact algorithm is not an approximate optimal solution but rather an optimal solution.

Heuristic algorithms: Heuristic algorithms can quickly yield acceptable solutions that satisfy the requirements within a reasonable timeframe. Dispatching rules are widely employed in scheduling problems [16–18]. In [16], Shanker and Tzen developed heuristic methods to minimize workload and balance the sequence of jobs in a flexible manufacturing system (FMS) and compared them with exact mixed-integer programming solutions. In [17], the dispatching rule outperforms swarm intelligence algorithms in a practical textile manufacturing system. In [18], Jun *et al.* addressed FJSP with release times, aiming to minimize the total weighted tardiness by learning dispatching rules.

In addition to dispatching rules, traditional heuristics are commonly employed in FJSP [19–22]. In [19], Chang *et al.* developed a beam search method for FMS with the objective of minimizing makespan. The proposed method yields a significantly shorter makespan compared to the commonly used shortest processing-time dispatching rule. In [20], Alvarez-Valdés *et al.* designed a heuristic algorithm specifically for FJSP in a glass factory, aiming to produce high-quality solutions in short computing times. In [21], Lee and Lee proposed novel heuristic functions for an algorithm based on the t-timed petri net. When minimizing makespan, these new heuristic functions demonstrated improved efficiency compared to previous functions in terms of the required number of states and computation time. In [22], Ziaee developed a heuristic based on a constructive procedure for the FJSP.

Several scholars have proposed hybrid heuristic methods to address the FJSP [23–27]. In [23], Xiong *et al.* presented a hybrid search algorithm for FMS that combines a heuristic best-first strategy with a controlled backtracking strategy. In [24], Mati and Xie proposed an integrated greedy heuristic for FJSP with more than two jobs. In [25], Mejía and Odrey proposed an algorithm that combines the A* search with an aggressive node-pruning strategy and improved evaluation functions to generate near-optimal schedules. In [26], Huang *et al.* introduced and evaluated a hybrid search strategy based on Petri nets for FMS scheduling, which combines the heuristic A* strat-

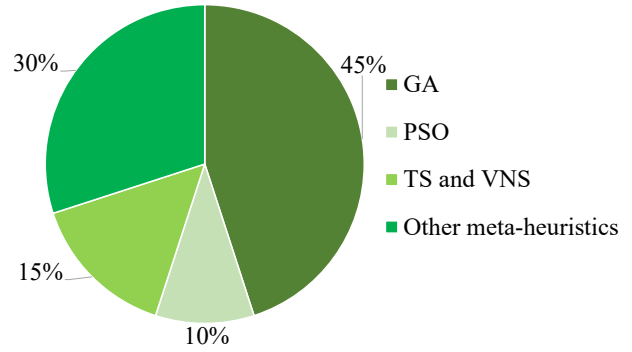


Fig. 2. Distribution of meta-heuristics for single objective FJSP.

egy with the DF strategy. In [27], Yuan and Xu proposed an integrated search heuristic that combines hybrid harmony search and large neighborhood search (LNS) to create a more powerful search mechanism for the FJSP.

Compared to exact algorithms, heuristic algorithms can solve FJSP more quickly. However, one disadvantage is that they may produce solutions of lower quality.

Meta-heuristics: Meta-heuristics can find solutions that meet the requirements within a short time. In this section, recent studies on meta-heuristics for classical FJSP are discussed. As seen in Fig. 2, among the meta-heuristic algorithms, genetic algorithm is the most frequently used, constituting 45% of the total. Population-based meta-heuristics constituting 85% of the total which includes GA, PSO and Jaya *et al.*

Population-based meta-heuristics follow a similar solution process. Firstly, different initialization strategies are utilized to initialize the population. Then, the objective function values of different individuals are calculated to evaluate their performance. A new solution is generated using a specific update method, and the objective function value of these solutions is used to determine whether to update. Finally, the iteration process is repeated until a stop criterion is met. Fig. 3 illustrates the general framework of population-based meta-heuristics.

Genetic algorithm (GA), initially proposed by Holland in 1992 [28], has been extensively utilized in FJSP. GA is a typical population-based meta-heuristics [29–37]. In [29], Pezzella *et al.* presented a GA that integrates different strategies for generating the initial population, selecting individuals for reproduction, and reproducing new individuals. In [30], Zhang *et al.* proposed an effective genetic algorithm that incorporates global selection (GS) and local selection (LS) to generate a high-quality initial population during the initialization stage. In [31], Driss *et al.* introduced a new genetic algorithm (NGA) that minimizes makespan in FJSP. This algorithm employs a new chromosome representation and incorporates various strategies for crossover and mutation. In [32], Li and Gao devel-

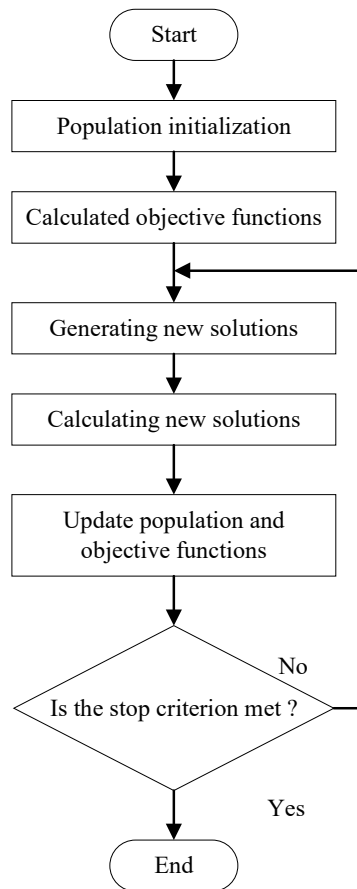


Fig. 3. General procedures framework of population-based meta-heuristics.

oped a hybrid algorithm (HA) by combining GA and tabu search (TS) to minimize makespan in FJSP. In [33], Chen *et al.* introduced a self-learning genetic algorithm (SLGA) for FJSP, which uses GA and adjusts its key parameters intelligently through reinforcement learning (RL). In [34], Shi *et al.* proposed a multi-population GA with an ER network (MPGA-ER) that exhibits significant improvements compared to traditional GA. In [35], Liu *et al.* presented a novel variable neighborhood descent hybrid genetic algorithm (VND-hGA) to overcome the low convergence speed and accuracy of GA. In [36], Park *et al.* developed a novel GA approach that utilizes multi-chromosomes for FJSP, addressing complex structural problems. Through numerical experiments, it was demonstrated that this u-COGA approach is effective in solving FJSPs. In [37], Sun *et al.* introduced a GA with variable neighborhood search (HGA-VNS) specifically designed for FJSP.

Particle swarm optimization (PSO), originally proposed by Kennedy based on bird behavior [38], is a population-based meta-heuristic algorithm that iteratively improves solutions. PSO learns from individual and population optimal solutions during the update process, providing fast search speed and high search accuracy, making it widely

used in FJSP. To address the issue of getting trapped in local optima, a quantum-behaved particle swarm optimization (QPSO) was proposed in [39], which incorporates a mutation operator and crossover operator. In [40], Nouri *et al.* proposed a multi-agent optimization system based on PSO to solve FJSP.

Artificial bee colony (ABC) algorithm, proposed by Karaboga, is a meta-heuristic algorithm inspired by the intelligent behavior of honeybees within a swarm [41]. In [42], Wang *et al.* presented an effective application of ABC algorithm. Numerical simulation results based on well-known benchmarks and comparisons with some existing algorithms demonstrate the effectiveness of the proposed ABC algorithm.

Trajectory-based metaheuristics are employed to solve FJSP problems by constructing and optimizing single solutions. These metaheuristics, such as TS and variable neighborhood search (VNS), are widely utilized in the field. The VNS, introduced by Mladenović and Hansen, is a meta-heuristic approach that systematically changes neighborhoods in the search space to find better-quality solutions [43]. Computational studies have demonstrated that this proposed algorithm is a viable and effective approach for addressing the FJSP. In [44], Yazdani *et al.* presented a parallel variable neighborhood search (PVNS) algorithm specifically designed for the FJSP. In [45], Zhang *et al.* developed a VNS based GA for the FJSP.

In recent years, various other population-based metaheuristics have been utilized to solve FJSP [47–52]. In [47], Xing *et al.* presented a knowledge-based ant colony optimization (KBACO) algorithm for FSJP. To minimize the makespan, a discrete grey wolf optimization (GWO) algorithm was developed in [48]. Other optimization algorithms that have been used include estimation of distribution algorithm (EDA) [49], Jaya algorithm [50], brain storm optimization algorithm (BSO) [51], dragonfly algorithm (DA) [52].

3.1.2 Approaches for multi-objective flexible job shop scheduling problem

Compared to single objective FJSP, multi-objective FJSP is characterized by increased complexity and difficulty in solution. Existing research literature primarily focuses on meta-heuristic methods, with limited use of exact algorithms and heuristic algorithms.

Fig. 4 illustrates the distribution of the three methods used for multi-objective FJSP. Meta-heuristic algorithms dominate, accounting for 90% of the total. In contrast to Fig. 1, the number of exact algorithms and heuristic algorithms has significantly decreased. This is because multi-objective FJSP involves a wider range of optimization objectives, making the solution more challenging. Accurate algorithms and heuristic algorithms struggle to meet these complex requirements.

The distribution of objective functions of multi-

Table 2. Objective function and approach of multi-objective FJSP.

Approach	Objective function	Number	Reference
Exact algorithms	Makespan, maximum workload, total workload	1	[53]
Heuristic algorithms	Makespan, maximum workload, total workload	2	[54,55]
Meta-heuristics	Makespan, maximum workload, total workload	14	[56-58,63-66,68-70,72,78,82,83]
	Makespan, total tardiness	2	[59,80]
	Makespan, total workload, total carbon emission	2	[71,76]
	Makespan, total energy consumption	2	[74,75]
	Makespan, energy efficiency, noise reduction	1	[60]
	Total carbon footprint, total late work criterion	1	[61]
	Total energy consumption, production cost	1	[62]
	Makespan, mean tardiness, mean flow time	1	[67]
	Makespan, energy cost	1	[73]
	Makespan, total energy cost, total labor cost, Maximal workload, total workload	1	[77]
	Maximum workload, total energy consumption	1	[79]
	Makespan, carbon footprints	1	[81]

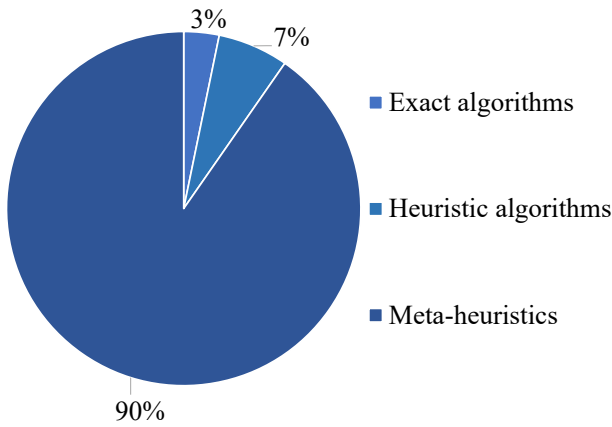


Fig. 4. Distribution of approaches for multi-objective FJSP.

objective FJSP is shown in Table 2. The most frequently occurring objective is the makespan, which appears in 28 multi-objective FJSP articles (90.3%). Apart from the makespan, the most common objectives are load-based, including maximal workload and total workload. Both maximal workload and total workload appeared 19 times (61.3%). Manufacturing enterprises face substantial economic pressure and environmental challenges due to energy consumption and its associated environmental impacts. Consequently, many scholars have been researching optimization objectives related to the environment, such as energy consumption and carbon emissions. The maximal workload and total workload are expressed by (10) and (11). Calculate total energy consumption, total tardiness, and mean tardiness by

$$E_t = \sum_{i=1}^n E_i, \quad (15)$$

$$T_t = \sum_{i=1}^n T_i, \quad (16)$$

$$\bar{T} = (\sum_{i=1}^n T_i)/n, \quad (17)$$

where E_i is the completion time of the job i . Other optimization objectives with fewer occurrences are not described in detail.

Exact algorithms and heuristic algorithms: Exact algorithms are widely recognized for their significant computational effort, especially when dealing with multi-objective problems (MOPs). Consequently, in the literature, exact algorithms are seldom employed to tackle MOPs. In [53], Soto *et al.* presented a novel multi-objective parallel branch-and-bound algorithm (MBB) which efficiently solved a set of instances of the multi-objective FJSP to optimality. This paper argues that MBB is a powerful choice to solve the small and medium-sized instances of multi-objective FJSP.

Several heuristic algorithms have been effectively applied to multi-objective FJSP [54,55]. In [54], Wang *et al.* developed a heuristic algorithm based on filtered beam search, enabling the efficient discovery of sub-optimal schedules for the multi-objective FJSP within a reasonable computational time. In [55], Pérez and Raupp introduced a hierarchical heuristic algorithm based on Newton's method to solve the multi-objective FJSP, which can be viewed as a continuous multi-objective unconstrained optimization problem.

Meta-heuristics: Fig. 5 illustrates the distribution of meta-heuristics for multi-objective FJSP. The majority of meta-heuristic algorithms are still genetic algorithms, representing about a quarter of the total. Then PSO, ABC and EA each accounted for 18%.

GA is also effectively applied to multi-objective FJSP,

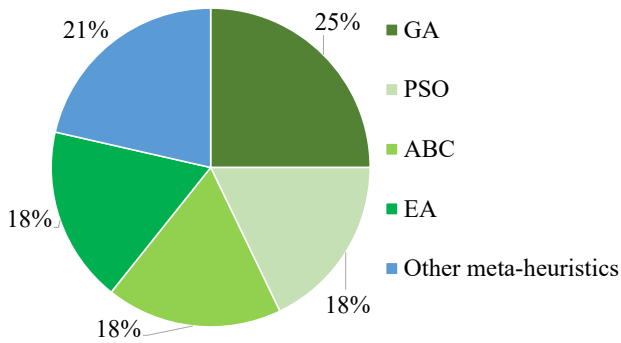


Fig. 5. Distribution of meta-heuristics for multi-objective FJSP.

where makespan, maximal machine workload, and total workload serve as common optimization criteria [56–59]. In [56], Gao *et al.* devised a hybrid genetic algorithm (HGA) that incorporates the variable neighborhood descent (VND) strategy, which includes two local search (LS) procedures: LS of moving one operation and LS of moving two operations. In [57], Wang *et al.* presented a multi-objective genetic algorithm (MOGA) based on immune and entropy principles. In [58], Bezoui *et al.* proposed a multi-objective genetic algorithm for FJSP. In [59], Türkylmaz *et al.* proposed an algorithm called bi-objective hybrid genetic algorithm with hypervolume contribution measure (BOHGA-HCM), which integrates GA with a multi-search algorithm for FJSP with objectives of minimizing makespan and total tardiness.

Given the increasingly pressing environmental issues, many scholars have started considering environmental factors such as energy consumption and carbon dioxide emissions in scheduling problems [60–62]. In [60], Yin *et al.* presented a MOGA based on a simplex lattice design for FJSP with three objectives: productivity, energy efficiency, and noise reduction. In [61], Piroozfard *et al.* introduced an improved multi-objective genetic algorithm for FJSP with objectives of minimizing total carbon footprint and total late work criterion. In [62], Wang *et al.* proposed a two-stage energy-saving optimization method for FJSP, where they applied a modified genetic algorithm (MGA) in the first stage and adopted a hybrid method that integrates GA with PSO in the second stage.

PSO is commonly applied to solving FJSP with objectives such as minimizing makespan, maximal machine workload, and total workload [63–67]. In [63], Zhang *et al.* proposed a hybrid algorithm that combines PSO with TS to solve the multi-objective FJSP. This hybrid algorithm demonstrates efficient and effective performance for large-scale FJSP. In [64], Moslehi and Mahnam proposed an integrated multi-objective approach that hybridizes PSO and LS for the multi-objective FJSP. In [65], Shao *et al.* introduced a hybrid algorithm that combines discrete particle swarm optimization (DPSO) and simulated annealing

(SA), where DPSO is employed for global search and SA is utilized for local search. In [66], Kato *et al.* presented a new approach that integrates PSO and random restart hill climbing (PSO + RRHC) for FJSP with three objectives. They developed novel strategies for population initialization, particle displacement, and stochastic allocation of operations. In [67], Singh *et al.* proposed a multi-objective particle swarm optimization (MOPSO) algorithm for FJSP with multiple objectives, aiming to improve efficiency and stability.

ABC is applied to solve multi-objective FJSP [68–72]. In [68], Li *et al.* proposed a hybrid Pareto-based discrete ABC algorithm to minimize makespan, maximal machine workload, and total workload in multi-objective FJSP. This study is the first to apply the discrete ABC algorithm and an external Pareto archive set to solve multi-objective FJSPs. In [69], Wang *et al.* presented the enhanced Pareto-based artificial bee colony (EPABC) algorithm for multi-objective FJSPs. With the aid of local search focusing on critical paths, the algorithm demonstrates effective local exploitation. In [70], Li *et al.* introduced the discrete artificial bee colony (DABC) algorithm, which presents a novel approach. Additionally in [71], Li *et al.* proposed the improved artificial bee colony (IABC) algorithm to tackle FJSP considering makespan, machine loading, and total carbon emissions as objectives. To address challenges such as blind searching, slow convergence, and unbalanced search directions, an efficient ABC algorithm was proposed in [72].

Evolutionary algorithms (EA) constitute most of the meta-heuristics approaches within the field and draw inspiration from classic evolution theory. These algorithms mimic evolutionary concepts such as reproduction, mutation, and selection to solve optimization tasks. Although the introduction of GA is already covered, subsequent approaches based on EA will be discussed. In [73], Wang introduced the hybrid multi-objective evolutionary algorithm based on decomposition (HMOEA/D). Multiple variations of non-dominated sorted genetic algorithm II (NSGA-II) has been employed to address multi-objective FJSP in [74,75]. For many-objective FJSP, non-dominated sorted genetic algorithm III (NSGA-III) and its variants were proposed by [76,77].

Hybridized tabu search algorithm (HTSA) for MOFJSP is proposed by Li *et al.* [78]. Hybridization is done by combining the VNS with TS meta-heuristics. The TS approach is used in the machine assignment subproblem to produce neighboring solutions and the VNS method is used in the operation scheduling subproblem to perform the local search for finding better solutions.

Many scholars use other meta-heuristic methods to solve the multi-objective FJSP [79–83]. In [79], Lei *et al.* applied the shuffled frog-leaping algorithm (SFLA) to solve FJSP. Lei *et al.* [80] proposed a two-phase meta-heuristic (TPM) based on imperialist competitive algo-

rithm (ICA) for multi-objective FJSP. In [81], Liu *et al.* employed fruit fly optimization algorithm (FOA) to solve it. A new hybrid discrete firefly algorithm (HDFA) is proposed by Karthikeyan *et al.* [82]. In [83], Yuan and Xu proposed a novel memetic algorithm (MA) specifically designed for multi-objective FJSP.

3.2. Dual-resource constrained

FJSP only takes into account machine constraints. In actual production, in addition to machine constraints, it is also necessary to consider worker constraints. Especially in the production of fine products such as aviation components, manual work must be integrated to complete product processing. Additionally, there are differences in the operation proficiency and working ability of different workers, resulting in varying efficiency when different workers operate the same equipment, which affects production efficiency. Assigning limited workers to a suitable machine for operation can improve operational stability.

3.2.1 Single objective dual-resource constrained flexible job shop scheduling problem

Compared with the classical FJSP, the dual-resource FJSP also considers the allocation problem of workers at the same time, which makes the solution more difficult. The existing research literature mainly focuses on meta-heuristic methods, and the use of exact algorithms and heuristic algorithms is limited. Makespan is the most common objective. All articles of the single objective DR-CFJSP take the makespan as the objective.

Berti *et al.* [84] proposed a heuristic method to assign operations to workers and machines. They measured the impact of rest allowance on system performance and provided management implications.

Meta-heuristics: GA is also effectively applied to DR-CFJSP [85-88]. In [85], Xianzhou and Zhenhe proposed a new immune GA for the DRCFJSP by combining the immune algorithm with the genetic algorithm (GA). This algorithm can avoid the premature convergence problem caused by high selective pressure. In [86], Peng *et al.* proposed a GA that adopts a new well-designed three-layer chromosome coding method and designed effective

crossover and mutation operators. The method is verified by case analysis. In [87], Wu *et al.* proposed a hybrid algorithm that combines GA and VNS. Computational experiments are carried out to evaluate the effectiveness of the proposed algorithm. The results show that the proposed algorithm can effectively and efficiently. In [88], Zhang *et al.* proposed a quantum genetic algorithm (QGA) to solve DRCFJSP with differences in operating time between operators.

In [89], Lie and Guo proposed a VNS algorithm for solving the DRCFJSP. In this algorithm, the solution to the problem is indicated as a quadruple string of the ordered operations and their resources. In [90], Yazdani *et al.* proposed two meta-heuristic algorithms that integrate SA and vibration damping optimization.

Many scholars use other meta-heuristic methods to solve the DRCFJSP [91-94]. In [91], Gong *et al.* introduced the hybrid artificial bee colony algorithm (HABCA). This algorithm incorporates effective encoding, decoding, crossover, and mutation operators. Zheng and Wang [92] proposed a knowledge-guided fruit fly optimization algorithm (KGFOA) to solve the DRCFJSP. In [93], Gao and Pan proposed a new algorithm called the two-vector representation of mixed multi-population micro-migratory bird optimization (SM2-MBO) algorithm. In [94], Madenoğlu and Baykasoğlu proposed weighted superposition attraction (WSA) for DRCFJSP. The results of the WSA algorithm are compared with those of the scheduling rule-based method and the greedy random adaptive search algorithm.

3.2.2 Multi-objective dual-resource constrained flexible job shop scheduling problem

The literature on the single-objective DRCFJSP primarily focuses on optimizing the makespan. The distribution of objective functions in MODRCFJSP is shown in Table 3. The makespan remains the most commonly considered objective function, appearing 9 times (90%). Following that, the load-related objective function, maximal machine workload, and total workload are also taken into account. Unlike the classical FJSP, MODRCFJSP incorporates indicators related to workers, such as maximum

Table 3. Objective function and approach of multi-objective MODRCFJSP.

Approach	Objective function	Number	Reference
Heuristic algorithms	Makespan, Standard deviation of makespan	1	[95]
Meta-heuristics	Makespan, maximum workload, total workload	3	[98,100,102]
	Makespan, production cost	1	[96]
	Makespan, human factor, green production	1	[97]
	Makespan, maximum worker workload	1	[99]
	Carbon emission, total cost of workers	1	[101]
	Makespan, total workload, product cost	1	[103]
	Makespan, maximum worker workload, total worker workload	1	[104]

worker fatigue and worker costs.

In order to minimize both the makespan and the standard deviation of the makespan, a multi-objective modified backtracking search algorithm (MBSA) framework was proposed [95]. MBSA incorporates an effective population initialization strategy, dynamic mutation and crossover operators, and a transfer criterion.

Meta-heuristics: Most of the methods for solving MODRCFJSP focus on GA. GA is also effectively applied to MODRCFJSP [96-99]. In [96], Liu *et al.* applied a Pareto-based hybrid GA to the bi-objective DRCFJSP, considering both the makespan and production cost. In [97], Gong *et al.* proposed a new hybrid GA to solve the DRCFJSP by integrating processing time, green production, and human factor indicators. In [98], Yazdani *et al.* proposed two multi-objective evolutionary algorithms for solving the MODRCFJSP: fast elite NSGA-II and non-dominated sorting genetic algorithm (NRGA). In [99], Tan *et al.* presented an enhanced NSGA-II (ENSGA) for the multi-objective fatigue-conscious DRCFJSP.

In [100], Gong *et al.* proposed a non-linear integer programming model for the multi-objective DRCFJSP. They also developed a MA to efficiently solve this problem. In [101], Zhu *et al.* introduced a MA for addressing the low-carbon FJSP, while considering worker learning. In [102], Lou *et al.* presented a MA based on learning and decomposition to tackle the multi-objective FJSP, taking into account the human factor.

When considering scheduling objectives such as makespan, machine workload, and product cost, a multi-objective evolutionary algorithm based on decomposition (MOEA/D) was presented [103]. Additionally, an improved differential evolution algorithm was introduced for optimizing the operation sequence. In [104], Shi *et al.* proposed a two-stage multi-objective PSO for the boredom-aware DRCFJSP.

3.3. Distributed flexible job shop scheduling problem

Intelligent production systems are of utmost importance in the realm of intelligent factories as they encompass the entire enterprise's production, scheduling, logistics, and management. With the advent of globalization, distributed manufacturing has gained prominence as a prevailing production mode, fueled by increased collaboration and corporate mergers and acquisitions. In the context of dis-

tributed FJSP, the allocation of each job to a specific manufacturing unit within a network of units becomes crucial. Each manufacturing unit encounters its own shop scheduling problem, which can be effectively represented as an FJSP. Consequently, the study of distributed FJSP holds immense significance.

3.3.1 Single objective distributed flexible job shop scheduling problem

In the literature on single objective distributed FJSP, the most commonly considered objective is the makespan, and the methods used often focus on GA.

Scholars have proposed both exact and heuristic approaches to solve the problem of distributed FJSP. For example, Meng *et al.* [105] developed a MILP model to optimize energy consumption in distributed FJSP. Ziaee [106] introduced a fast heuristic algorithm based on a constructive process, which enables the quick generation of high-quality plans for distributed FJSP.

GA are widely utilized in solving distributed FJSP [107-111]. In earlier work [107,108], Chan *et al.* studied distributed FJSP and proposed a GA that employs the concept of dominant genes to solve the problem. Giovanni and Pezzella [109] presented an improved GA for resolving distributed FJSP. This algorithm incorporates a novel local search-based operator, which enhances the quality of solutions by refining the most promising individuals in each generation. Chang and Liu [110] introduced a hybrid genetic algorithm (HGA) for distributed FJSP. The HGA incorporates various crossover and mutation operators, aiming to increase the likelihood of finding the optimal solution. In [111], Lu *et al.* proposed a GA_JS for solving distributed FJSP.

3.3.2 Multi-objective distributed flexible job shop scheduling problem

In the literature on multi-objective distributed FJSP, meta-heuristic algorithms are commonly utilized. As shown in Table 4, all articles take the makespan as the objective. However, it is important to note that these studies also take into account other significant optimization objectives, such as total energy consumption, maximum workload, carbon emission, and manufacturing quality.

In [112], Xu *et al.* proposed a HGA and TS algorithm, named H-GA-TS, which aims to minimize not only

Table 4. Objective function and algorithm of multi-objective distributed FJSP.

Meta-heuristic	Objective function	Number	Reference
MA	Makespan, maximum workload, total energy consumption	1	[113]
	Makespan, total energy consumption, delay time, manufacturing quality	1	[114]
GA	Makespan, costs, quality, carbon emission	1	[112]
EDA	Makespan, maximal workload, total workload, tardiness	1	[115]
SA	Makespan, total energy consumption	1	[116]

the makespan but also costs, quality, and carbon emissions. Similarly, in [113], Luo *et al.* developed an efficient memetic algorithm (MA) specifically designed for solving distributed FJSP problems with transfers. Additionally, Sang and Tan introduced a high-dimensional many-objective memetic algorithm (HMOMA) in [114]. In [115], Li *et al.* presented a hybrid Pareto-based TS algorithm for the distributed FJSP. Moreover, Du *et al.* proposed a hybrid algorithm in [116] that combines the EDA and VNS techniques to solve the distributed FJSP, considering both makespan and total energy consumption.

4. DYNAMIC FLEXIBLE JOB SHOP SCHEDULING

Dynamic events can be classified into two primary categories: resource-related and job-related. Resource-related events encompass disruptions or changes in the availability or condition of production resources, such as machines, tools, or materials. These events often require adjustments in resource allocation, rescheduling of operations, or resource substitutions. Table 5 gives the common centralised dynamic events. On the other hand, job-related events directly impact the production orders or jobs themselves, including changes in due dates, priority adjustments, alterations in processing sequences, or modifications in product specifications. Effective dynamic scheduling algorithms and techniques are essential in promptly evaluating the impact of these events on the existing schedule, identifying necessary adjustments, and implementing them to minimize disruptions and optimize resource utilization. Fig. 6 illustrates the five common types of dynamic events. Researchers usually consider dynamic events in a mixed manner. If a single dynamic event is discussed separately, it will form cross repetition. Therefore, this article divides the literature related to dynamic scheduling into single dynamic event and multiple dynamic events.

4.1. Resource-related dynamic events

In the actual production process, machines, tools, materials and workers are all resource-related and they are subject to changes in the state of manufacture at any time, thus causing dynamic events. Unfortunately, in almost all the literature on dynamic scheduling, researchers have fo-



Fig. 6. DFJSP Categories.

Table 6. Different algorithms for handling breakdown.

Intelligent swarm algorithm	Number	Reference
GA	5	[117-121]
NSGA-II	3	[122-124]
PSO	2	[125,126]
TLO	2	[128,129]
Jaya	2	[130,131]
Migrating bird algorithm	1	[127]
Dispatching rules	1	[132]

cused only on the dynamic handling of machine failure classes and have not considered the remaining resource-related events. Therefore, in this chapter we will focus on machine failures as a class of dynamic disturbances.

Table 6 shows the different algorithms for handling breakdowns. From a methodological perspective, intelligent population algorithms have emerged as a prevalent approach for addressing single machine failure problems. Since 2009, a multitude of meta-heuristic algorithms have been extensively utilized to tackle dynamic flexible job shop scheduling problem (DFJSP), owing to their straightforward structure, and most importantly, their ability to generate high-quality solutions.

A simulation-based GA that effectively handles simulated random faults is proposed in [117]. By incorporating right-shifting scheduling and rescheduling techniques, they successfully determine the optimal solution and make it a seminal method referenced by subsequent researchers. A two-stage hybrid GA, the first stage of the algorithm focuses on generating pre-scheduling, while in the second stage, it takes machine faults into account during the decoding process in [118]. Numerous enhancements have been made in [119-121]. These enhancements encompass various aspects, including initialization, search strategy, and other relevant factors. While these advancements are crucial for addressing the challenges posed by DFJSP, it is

Table 5. DFJSP considering different dynamic events.

Dynamic type	Dynamic events	Number	Reference
Resource-related	Machine breakdown	16	[117-132]
Job-related	Job insertion	4	[133-136]
	Process time change	7	[137-143]
Multiple	Different combinations	23	[144-166]

important to note that GA often exhibit slow convergence in multi-objective optimization scenarios. Therefore, the development of algorithms specifically tailored for this problem becomes imperative.

In [122-124], they employ an improved version of the NSGA-II algorithm to solve the single machine breakdown problem. These researchers have not only enhanced the traditional GA within their frameworks, but they have also incorporated factors such as the production state of the machine, the probability of machine failure, and the floating-point time of machine failure. Their approaches have demonstrated superior performance when compared to advanced algorithms developed in recent years.

In addition to advances in GA, in [125,126], they solve single machine fault problem with improved PSO. In [127], they improve search ability based on migrating bird algorithm and compared with advanced algorithms in recent years, it has been proven to have good performance.

In [128,129], they use teacher learning optimization algorithm to find the optimal solution in a reasonable time and demonstrate higher performance than GA and PSO. In [130,131], they utilize the almost parameterless meta-heuristic Jaya algorithm to improve the initialization strategy and local search method. In [132], they use gene expression programming rules to extract scheduling rules. However, it was observed that the optimization process associated with this approach is significantly more complex when compared to GA.

4.2. Job-related dynamic events

Job-related dynamic events encompass job insertion, process time change, due date change, and job uncertainty. Among these events, job insertion and process time change have garnered significant attention from researchers. This paper serves as a comprehensive summary of the published articles on these topics.

4.2.1 Job insertion

Job insertion occurs when a stochastic order is received or when there is an urgent need for machining a job, necessitating proactive rescheduling. The essence of job insertion lies in efficiently integrating novel orders into the incumbent schedule to ensure optimal stability and progress in the realm of FJSP and intelligent manufacturing.

In [133], they propose a GA based matching rescheduling algorithm, comparing the no-rescheduling strategy with the full rescheduling strategy for newly arrived orders, where the no-rescheduling strategy has higher stability. In [134], they propose a reactive heuristic algorithm that takes into account job shop utilisation, due time tightness and job shop flexibility. In [135], they propose a two-stage ABC to solve the scheduling and rescheduling of new orders under the constraint of manufacturing return uncertainty. In [136], they propose an online rescheduling

framework based on deep Q-networks for new job insertions and assigns them to machines for execution based on a combination of six scheduling rules, which has effectiveness and generality, but requires a large amount of actual workshop data.

In general, dynamic scheduling for artifact insertion alone is still relatively rare. We believe that artefact insertion should weigh the advantages and disadvantages of inserting new artefacts after their arrival into the pre and post rescheduling scenarios, and determine the priority of existing artefacts versus arriving artefacts.

4.2.2 Process time change

Fixed processing events may not meet the requirements of the actual production environment, and the processing time is usually within a certain interval, which can be represented by a generalized fuzzy number. Usually, this type of problem is attributed to fuzzy FJSP. This type of problem is similar to the handling method of FJSP, where the processing time is fuzzified during the modeling process.

In [137], they proposed an improved GA combining TS and heuristic seeding, which was simulated on deterministic and fuzzy instances to show the synergy between different components of the hybrid algorithm. In [138], they transformed the decoding method into a feasible scheduling solution in a fuzzy sense, combining LS with teaching and learning mechanisms. In [139], they defined the uncertain processing time as interval processing time, and the corresponding completion time is also gray. The designed elite retention GA is suitable for solving such problems. In [140], they proposed a PSO and Cauchy distribution, he proposed an improved GA combining tabu search and heuristic seeding, which was simulated on deterministic and fuzzy instances to show the synergy between different components of the hybrid algorithm. In [141], they transformed the decoding method into a feasible scheduling solution in a fuzzy sense, combining local search with teaching and learning mechanisms. In [142], they defined the uncertain processing time as interval processing time, and the corresponding completion time is also gray. The designed elite retention GA is suitable for solving such problems. In [143], they believe that blurring time can bring many uncertain factors, such as blurring the total energy consumption. A two population evolutionary algorithm with feedback is proposed to solve this kind of dynamic scheduling problem.

4.3. Multiple dynamic events

Fig. 7 illustrates the main contributions to the development of dynamic scheduling and is available for the reader to review. Methodologically, the processing methods that consider multiple dynamic events are mostly intelligent swarm algorithms [144-163], with some deep learning and neural networks [164-166]. To investigate hybrid dynamic events, it is essential to consolidate them based on their

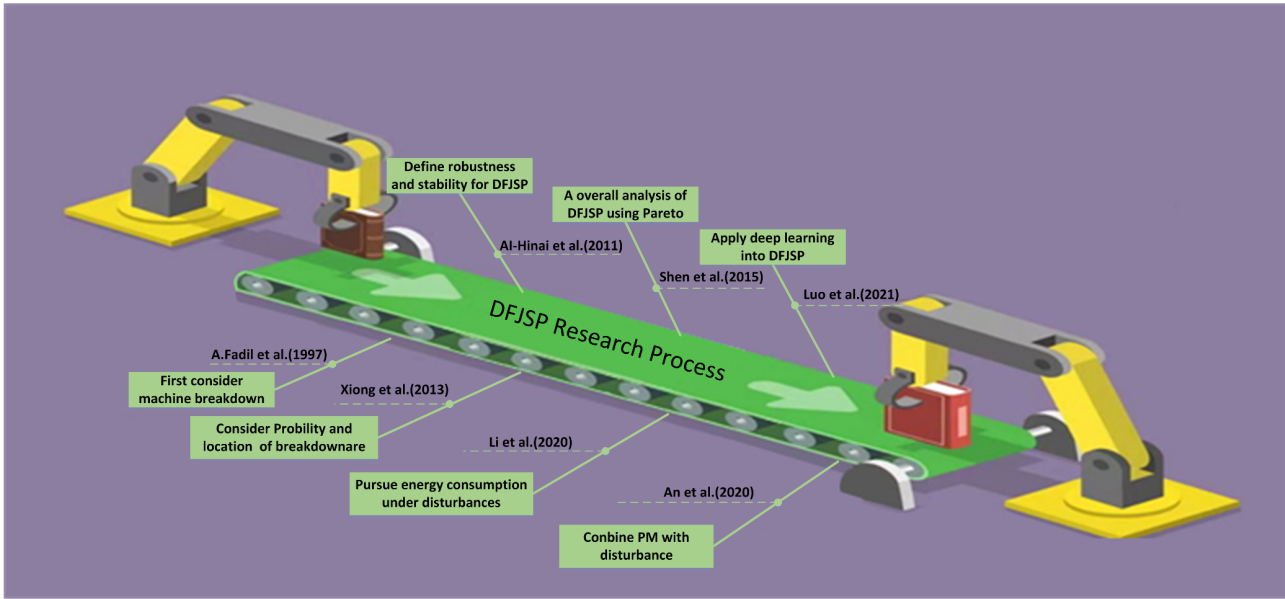


Fig. 7. DFJSP research progress.

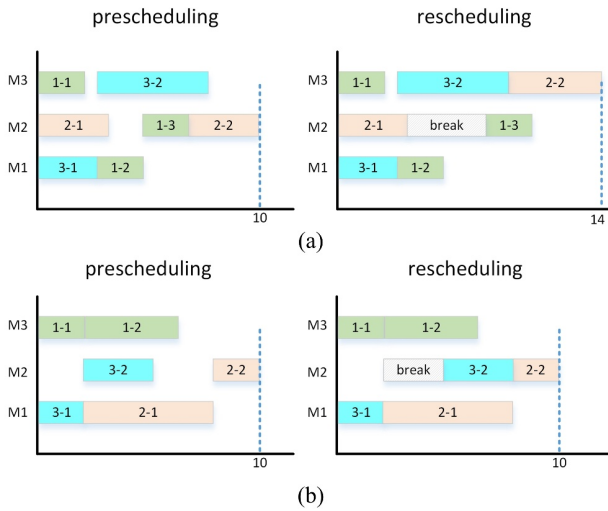


Fig. 8. Example of schedule of robustness and stability.

shared concerns in the context of workshop scheduling and intelligent manufacturing.

Robustness and stability are a reflection of the ability to handle well under dynamic events. More intuitively, a robust and stable scheduling solution is one in which there is little change in resources and artefacts after a dynamic event. Robust and stable systems reduce the cost of re-scheduling and ensure a seamless transition between pre-scheduled and re-scheduled solutions. As an example in Fig. 8, the shaded area is the event corresponding to the failure of the machine, and schedule (b) is more robust and stable than schedule (a). In [118,145], they define it as the difference between prescheduling and rescheduling. In

[119], they use the same definition of robustness, but the operation time before the post-fault period has been determined, so it defines stability as the difference between the expected and actual operation time of the outstanding operation. In [122], they use Monte Carlo methods to approximate the predicted value of the optimization objective under machine failure, thus measuring the robustness of the scheduling solution. In [127], they define the stability indicator as the weighted sum of the maximum completion time and the relative deviation of the machine displacement. In [121,123,124,129,148,154,156,164,166], they almost follow method of definition by [10]. The interesting thing we find that as energy consumption and environmental issues become increasingly important, more and more research is focused on optimising energy consumption in [26,27,149,151,160,163].

Few studies have achieved real-time scheduling. A hybrid rescheduling strategy combining active and passive optimization models with rescheduling addresses the cost of relationships in manufacturing in [158]. For the first time, in [161], they combine rescheduling with process planning to achieve energy-efficient production and process flexibility on the shop floor, and transformed the problem into a mixed-integer planning model to optimise energy consumption for flexible manufacturing.

In [156], they build an online rescheduling framework that uses deep learning to solve DFJSP, where at each rescheduling point, the input is an optimisation goal for the current state and four different forms of reward functions are used to optimise tardiness and machine utilisation after selecting the corresponding behaviour. Different scheduling rules are assigned to different machines, improving the efficiency of scheduling. Deep learning is

invoked to provide new directions for solving DFJSP. In [159], they use two deep learning network optimisation scheduling schemes, improved local search and combinatorial ideas, he verified performance improvements of 57%, 164% and 28% for completion time, machine utilisation and processing latency respectively.

In [166], they solve common dynamic problems based on hierarchical multi-intelligence deep reinforcement learning, where the target as controller needs to determine the optimization object, the task and the machine as the lower actuator, and different job selection rules and machine assignment rules increase shop floor efficiency. In [164], they propose an improved imperialist competition algorithm to generate training data and use convolutional neural networks to train scheduling models that can quickly evaluate the robustness of scheduling. The following diagram shows the references that have contributed most to the development of DFJSP in terms of methods or problems, and we have listed them to sort out the development of near-dynamic workshop scheduling.

5. PREVENTIVE MAINTENANCE

Fault prediction and health management is a comprehensive fault detection, isolation prediction and health state management technology. In FJSP, preventive maintenance is widely used for equipment health management. Preventive maintenance is clearly necessary in industrial production to prevent failures through regular maintenance and repair of equipment, which is critical in low accident production such as aerospace. PM is often expressed as a maintenance strategy on limited or unlimited resources to optimize dispatch objectives. In almost all PM studies, maintenance strategies can be broadly divided into two categories, and in [76], they define them as two types: fixed periods and fixed windows. Fixed periods are maintenance policy where the maintenance process has a predetermined start time, while fixed windows allow maintenance to be carried out within a window of time. In addition to this, maintenance based on machine condition (CBM) has also been some research developments.

Table 7 collates the main PM applications in FJSP from three major publishers Springer, Science Direct and IEEE containing three main areas: maintenance strategy. Of all the 24 FJSP articles counted that included PM operations, 12.5% used fixed periods maintenance strategy only, 8.3%

considered and compared fixed periods and fixed windows in combination. The difference between the two policies is that fixed periods policy requires all maintenances to start at the same time, while fixed windows of maintenance can be slid over a window of time from the earliest time to the latest time. The rest are using fixed windows strategy.

5.1. Fixed periods

Research focus on single machine scheduling and flow shop scheduling with PM, in [168] they firstly propose the consideration of machine interval maintenance in FJSP and used a hybrid GA to solve the FJSP with machine availability period. In [169], they enriched their previous experiments with two areas that FJSP should focus on in the future: production and maintenance, corresponding to makespan and maintenance costs respectively. PM can significantly influence FJSP, preventive maintenance and learnable genetic architecture can significantly improve the quality of the solution. In [170], they have in common that PM activities are established in advance, so PM is not scheduled as a production constraint.

5.2. Fixed windows

In [171], they consider the variable availability constraints and completion time of maintenance tasks in FJSP. In his paper, both factors need to be determined during the scheduling process. In [173], they build a reliability model for maintenance activities, deciding when to perform PM activities while optimizing completion times. The correct PM policy selection for the operations and production scheduling becomes the key difficulty, they also point out that the recoverability of jobs can also be taken into account. The earliest multi-objective FJSP with parallel machine and maintenance constraints included found in the literature are small-scale, medium-scale, and large-scale problems in [174]. In [172,175-179], they publish a similar further study. However, the above study cost a lot of maintenance resources and full coverage of the machine for preventive maintenance will inevitably lead to longer production times.

With the development of smart manufacturing, there is a growing interest in production scheduling, especially the unavailability periods in scheduling problems. In [180], they propose a novel approach to deal with FJSP with nonpreemptive unavailable environment for PM to minimize the makespan, the joint planning maintenance policy adopted can handle random situations such as machine breakdowns. Each machine has a different failure rate and its optimum age of repair, i.e., the optimum threshold, needs to be taken into account when carrying out repairs and replacements. In addition to PM, in [181] they propose to consider a corrective maintenance strategy, where the system can be monitored and managed based on the condition of the equipment through the detection of various sensors. In [182], they believe that if the machine ex-

Table 7. Different algorithms for handling breakdown.

PM policy	Number	Reference
Fixed periods	3	[168-170]
Fixed windows	16	[162,171-185]
Others	5	[186-190]

ceeds its optimal threshold then preventive maintenance or replacement before processing will improve the expected completion time of the job, and they also take into account the carbon emission to achieve green shop scheduling. Combining maintenance planning with machining planning increases machine availability and production scheduling robustness in [183]. They integrate maintenance planning with the transportation process and evaluate the proposed method by general distance (GD) and spread distance (SD), and experiments show that their method is effective in solving the integrated scheduling problem. They devote unlimited maintenance resources to machine restoration, which is clearly limiting. In [184], they define their work as flexible production combined with flexible maintenance scheduling, taking into account the availability of maintenance time windows, and they propose several different models that show their superiority and effectiveness. In [185], they carry out simulations on actual vehicle production data, results shows that joint job scheduling and maintenance job allocation was required to obtain a maximum completion time similar to the maximum completion time, without taking maintenance into account. Non-complex scheduling with a maintenance window can improve the maximum completion time by more than 20 percentage points compared to complex scheduling. Furthermore, only a limited number of maintenance activities can be compensated for and that higher unavailability times will inevitably lead to a negative impact on the maximum completion time unless the capacity of maintenance workers is improved.

5.3. Others

We divide the others into condition-based maintenance (CBM) and compare the effectiveness of the two methods of maintenance.

5.3.1 Condition-based maintenance

In order to reduce the unavailability of machines and maintain the reliability of the system, the condition based maintenance protect the machines depending on the detection status of the machine, the application of CBM leads FJSP dynamic and more effective in [186,187]. In [188], they analyse the shortcomings of the imperialist algorithm, combined it with a simulated annealing algorithm, and modelled a strategy based on conditional maintenance that improves the efficiency of maintenance.

5.3.2 Fixed windows and fixed periods

A few papers compare the two approaches and analyse their advantages and disadvantages.

In [189], they consider the contrast between two maintenance strategies with unlimited maintenance resources and limited maintenance resources. PM with infinite maintenance resources is not constrained, while PM with only one maintenance resource is resource constrained. In other

words, there is no constraint on PM in the infinite maintenance resource case, while there is a resource constraint on PMs with only one maintenance resource. They experimentally demonstrated that the scheduling solution under the fixed period policy, whether the PM policy was carried out from the earliest time or executed at the latest time, resulted in a longer completion time than the fixed window policy. Considering both on time and late deliveries, in [190] they conclude from three sets of experiments that the mean tardiness, makespan and mean flow time of a fixed windows outperforms the fixed periods policy, regardless of how the time of the fixed window is set. In general, fixed-windows maintenance strategy is preferable to fixed-periods strategy, but the difficulty with fixed-window maintenance is how to determine the timing of repairs so that they can be flexible. They point out that the biggest disadvantage of the PM strategy is that it leads to a reduction in machine availability or machine downtime. Overall, preventive maintenance needs to be as efficient as possible with as little impact as possible to ensure that failures are effectively avoided.

In general, preventive maintenance still requires extensive research in the field of production scheduling. Maintenance strategies based on equipment condition and data-driven approaches [191-193] are currently the approaches to address preventive maintenance.

6. CONCLUSIONS AND FUTURE RESEARCH DIRECTIONS

6.1. Conclusions

This review summarises flexible job shop scheduling problems: optimization objectives, practical constraints and dynamic scheduling under disturbances are the focus of attention, with emphasis on the handling of machine failures as unexpected disturbances. By analyzing the existing literature, several intriguing conclusions can be drawn as follows:

Objectives and constraints: Although the makespan is the most studied criterion in the literature, it seems that it is rarely relevant in practical settings. On the other hand, considering multiple criteria is often necessary. The classical FJSP is an academic optimization problem to solve practical scheduling problem.

Dynamic events: Researchers are more interested in rescheduling strategies than preventive maintenance. Machine breakdowns and job insertions are the most studied dynamic events and considering more disturbances and constraints is harder to deal with, but it is close to the practical settings.

Methods and models: A large number of swarm intelligence algorithms have been developed to address the FJSP. Additionally, intelligent population algorithms have matured in solving DFJSP, deep reinforcement learning and neural networks provide new directions.

6.2. Future research directions

We believe that future research should focus on the following directions:

Conduct more experiments to specific industries:

Production scheduling radiates to many industries, including semiconductor, medical, military, etc. The current simulations are mainly based on publicly available data sets, but future research should be specific to different industries in order to analyse their characteristics. By incorporating supplementary constraints, the extended FJSP can more accurately reflect actual production.

Dynamic scheduling still needs a lot of research: Dynamic scheduling is still predominantly event-driven, and we believe that realising cloud-side information interaction and using real-time information to reschedule is in line with smart manufacturing requirements. The digital twin factory, which combines virtual and real production, is bound to be an important direction for future research. More dynamic time should be taken into consideration, and adaptive methods may solve this kind of interference problem [194,195]. In addition, future work can optimize the triggering form such as sample based dynamic event triggering or time threshold to simulate the actual production environment accurately [196,197].

Apply fault diagnosis to dynamic scheduling: The occurrence of faults and disturbances will seriously affect the system operation status, and the faults are relatively simple and assumed to be known, without considering the problem of fault diagnosis [198-200], which will be combined with production line scheduling to study the problem of intelligent production line fault diagnosis and propagation, and realize intelligent scheduling and fault-tolerant operation and maintenance of production lines [201-203].

CONFLICTS OF INTERESTS

The authors declare that there is no competing financial interest or personal relationship that could have appeared to influence the work reported in this paper. Bin Jiang currently serves as a Senior Editor for International Journal of Control, Automation, and Systems.

REFERENCES

- [1] M. J. Triebe, F. Zhao, and J. W. Sutherland, "Genetic optimization for the design of a machine tool slide table for reduced energy consumption," *Journal of Manufacturing Science and Engineering-Transactions of the ASME*, vol. 143, no. 10, pp. 101003, 2021.
- [2] W. Xia and Z. Wu, "An effective hybrid optimization approach for multi-objective flexible job-shop scheduling problems," *Computers & Industrial Engineering*, vol. 48, no. 2, pp. 409-425, 2005.
- [3] J. F. Gonçalves, J. J. de Magalhães Mendes, and M. G. C. Resende, "A hybrid GA for the job shop scheduling problem," *European Journal of Operational Research*, vol. 167, no. 1, pp. 77-95, 2005.
- [4] B. Çaliş and S. Bulkan, "A research survey: review of AI solution strategies of job shop scheduling problem," *Journal of Intelligent Manufacturing*, vol. 26, pp. 961-973, 2015.
- [5] P. Brandimarte, "Routing and scheduling in a flexible job shop by tabu search," *Annals of Operations research*, vol. 41, no. 3, pp. 157-183, 1993.
- [6] J. Hurink, B. Jurisch, and M. Thole, "Tabu search for the job-shop scheduling problem with multi-purpose machines," *Operations Research Spektrum*, vol. 15, pp. 205-215, 1994.
- [7] S. Dauzère-Pérés and J. Paulli, "An integrated approach for modeling and solving the general multiprocessor job-shop scheduling problem using tabu search," *Annals of Operations research*, vol. 70, no. 0, pp. 281-306, 1997.
- [8] S. Q. Liu and E. Kozan, "Scheduling trains as a blocking parallel-machine job shop scheduling problem," *Computers & Operations Research*, vol. 36, no. 10, pp. 2840-2852, 2009.
- [9] C. Özgüven, L. Özbakır, and Y. Yavuz, "Mathematical models for job-shop scheduling problems with routing and process plan flexibility," *Applied Mathematical Modelling*, vol. 34, no. 6, pp. 1539-1548, 2010.
- [10] E. M. A. E. Abd, S. A. O. Mohamed, and B. A. H. Mohamed, "Optimality of the flexible job shop scheduling problem," *African Journal of Mathematics and Computer Science Research*, vol. 4, no. 10, pp. 321-328, 2011.
- [11] V. Roshanaei, A. Azab, and H. ElMaraghy, "Mathematical modelling and a meta-heuristic for flexible job shop scheduling," *International Journal of Production Research*, vol. 51, no. 20, pp. 6247-6274, 2013.
- [12] L. Huang and R. Su, "An auto-MILP model for flexible job shop scheduling problem," *IFAC-PapersOnLine*, vol. 55, no. 3, pp. 137-142, 2022.
- [13] M. C. Zhou, H. S. Chiu, and H. H. Xiong, "Petri net scheduling of FMS using branch and bound method," *Proceedings of IECON'95-21st Annual Conference on IEEE Industrial Electronics*, pp. 211-216, 1995.
- [14] S. Lloyd, H. Yu, and N. Konstantas, "FMS scheduling using Petri net modeling and a branch & bound search," *Proceedings. IEEE International Symposium on Assembly and Task Planning*, pp. 141-146, 1995.
- [15] R. S. Hansmann, T. Rieger, and U. T. Zimmermann, "Flexible job shop scheduling with blockages," *Mathematical Methods of Operations Research*, vol. 79, pp. 135-161, 2014.
- [16] K. Shanker and Y. J. J. Tzen, "A loading and dispatching problem in a random flexible manufacturing system," *International Journal of Production Research*, vol. 23, no. 3, pp. 579-595, 1985.

- [17] M. A. Ortíz, L. E. Betancourt, K. P. Negrete, F. De Felice, and A. Petrillo, "Dispatching algorithm for production programming of flexible job-shop systems in the smart factory industry," *Production Planning & Control*, vol. 264, pp. 409-433, 2018.
- [18] S. Jun, S. Lee, and H. Chun, "Learning dispatching rules using random forest in flexible job shop scheduling problems," *International Journal of Production Research*, vol. 57, no. 10, pp. 3290-3310, 2019.
- [19] Y. L. Chang, H. Matsuo, and R. S. SULLIVAN, "A bottleneck-based beam search for job scheduling in a flexible manufacturing system," *The International Journal of Production Research*, vol. 27, no. 11, pp. 1949-1961, 1989.
- [20] R. Alvarez-Valdés, A. Fuertes, J. M. Tamarit, G. Giménez, and R. Ramos, "A heuristic to schedule flexible job-shop in a glass factory," *European Journal of Operational Research*, vol. 165, no. 2, pp. 525-534, 2005.
- [21] J. Lee and J. S. Lee, "Heuristic search for scheduling flexible manufacturing systems using lower bound reachability matrix," *Computers & Industrial Engineering*, vol. 59, no. 4, pp. 799-806, 2010.
- [22] M. Ziaee, "A heuristic algorithm for solving flexible job shop scheduling problem," *The International Journal of Advanced Manufacturing Technology*, vol. 71, pp. 519-528, 2014.
- [23] H. H. Xiong, M. C. Zhou, and R. J. Caudill, "A hybrid heuristic search algorithm for scheduling flexible manufacturing systems," *Proceedings of IEEE International Conference on Robotics and Automation*, pp. 519-528, 2016.
- [24] Y. Mati, N. Rezg, and X. Xie, "An integrated greedy heuristic for a flexible job shop scheduling problem," *Proc. of IEEE International Conference on Systems, Man and Cybernetics*, pp. 2534-2539, 2001.
- [25] G. Mejía and N. G. Odrey, "An approach using Petri nets and improved heuristic search for manufacturing system scheduling," *Journal of Manufacturing Systems*, vol. 24, no. 2, pp. 79-92, 2005.
- [26] B. Huang, Y. Sun, and Y. M. Sun, "Scheduling of flexible manufacturing systems based on Petri nets and hybrid heuristic search," *International Journal of Production Research*, vol. 46, no. 16, pp. 4553-4565, 2008.
- [27] Y. Yuan and H. Xu, "An integrated search heuristic for large-scale flexible job shop scheduling problems," *Computers & Operations Research*, vol. 40, no. 12, pp. 2864-2877, 2013.
- [28] J. H. Holland, *Adaptation in Natural and Artificial Systems: An Introductory Analysis With Applications to Biology, Control, and Artificial Intelligence*, Cambridge, MA, 1992.
- [29] F. Pezzella, G. Morganti, and G. Ciaschetti, "A genetic algorithm for the flexible job-shop scheduling problem," *Computers & operations research*, vol. 35, no. 10, pp. 3202-3212, 2008.
- [30] G. Zhang, L. Gao, and Y. Shi, "A genetic algorithm for the flexible job-shop scheduling problem," *Computers & operations research*, vol. 35, no. 10, pp. 3202-3212, 2008.
- [31] I. Driss, K. N. Mouss, and A. Laggoun, "A new genetic algorithm for flexible job-shop scheduling problems," *Journal of Mechanical Science and Technology*, vol. 29, pp. 1273-1281, 2015.
- [32] X. Li and L. Gao, "An effective hybrid genetic algorithm and tabu search for flexible job shop scheduling problem," *International Journal of Production Economics*, vol. 174, pp. 93-110, 2016.
- [33] R. Chen, B. Yang, S. Li, and S. Wang, "A self-learning genetic algorithm based on reinforcement learning for flexible job-shop scheduling problem," *Computers & Industrial Engineering*, vol. 149, pp. 106778, 2020.
- [34] X. Shi, W. Long, Y. Li, and D. Deng, "Multi-population genetic algorithm with ER network for solving flexible job shop scheduling problems," *PloS one*, vol. 15, no. 10, pp. 0233759, 2020.
- [35] Z. Liu, J. Wang, C. Zhang, H. Chu, G. Ding, and L. Zhang, "A hybrid genetic-particle swarm algorithm based on multilevel neighbourhood structure for flexible job shop scheduling problem," *Computers & Operations Research*, vol. 135, pp. 105431, 2021.
- [36] J. S. Park, H. Y. Ng, T. J. Chua, Y. T. Ng, and J. W. Kim, "Unified genetic algorithm approach for solving flexible job-shop scheduling problem," *Applied Sciences*, vol. 11, no. 14, pp. 6454, 2021.
- [37] K. Sun, D. Zheng, H. Song, Z. Cheng, X. Lang, W. Yuan, and J. Wang, "Hybrid genetic algorithm with variable neighborhood search for flexible job shop scheduling problem in a machining system," *Expert Systems with Applications*, vol. 215, pp. 119359, 2023.
- [38] J. Kennedy and R. Eberhart, "Particle swarm optimization," *Proc. of ICNN'95-international conference on neural networks*, pp. 1942-1948, 1995.
- [39] M. R. Singh and S. S. Mahapatra, "A quantum behaved particle swarm optimization for flexible job shop scheduling," *Computers & Industrial Engineering*, vol. 93, pp. 36-44, 2016.
- [40] M. Nouri, A. Bekrar, A. Jemai, S. Niar, and A. C. Ammari, "An effective and distributed particle swarm optimization algorithm for flexible job-shop scheduling problem," *Journal of Intelligent Manufacturing*, vol. 29, pp. 603-615, 2018.
- [41] D. Karaboga, "An idea based on honey bee swarm for numerical optimization," *Technical Report-tr06, Erciyes University, Engineering Faculty, Computer Engineering department*, vol. 200, pp. 1-10, 2005.
- [42] L. Wang, G. Zhou, Y. Xu, S. Wang, and M. Liu, "An effective artificial bee colony algorithm for the flexible job-shop scheduling problem," *International Journal of Advanced Manufacturing Technology*, vol. 60, pp. 303-315, 2012.
- [43] N. Mladenović and P. Hansen, "Variable neighborhood search," *Computers & operations research*, vol. 24, no. 11, pp. 1097-1100, 1997.
- [44] M. Yazdani, M. Amiri, and M. Zandieh, "Flexible job-shop scheduling with parallel variable neighborhood search algorithm," *Expert Systems with Applications*, vol. 37, no. 1, pp. 678-687, 2010.

- [45] G. Zhang, L. Zhang, X. Song, Y. Wang, and C. Zhou, "A variable neighborhood search based genetic algorithm for flexible job shop scheduling problem," *Cluster Computing*, vol. 22, pp. 11561-11572, 2019.
- [46] F. Glover, "Future paths for integer programming and links to artificial intelligence," *Computers & Operations Research*, vol. 13, no. 5, pp. 533-549, 1986.
- [47] L. N. Xing, Y. W. Chen, P. Wang, Q. S. Zhao, and J. Xiong, "A knowledge-based ant colony optimization for flexible job shop scheduling problems," *Applied Soft Computing*, vol. 10, no. 3, pp. 888-896, 2010.
- [48] T. Jiang and Z. Chang, "Application of grey wolf optimization for solving combinatorial problems: Job shop and flexible job shop scheduling cases," *IEEE Access*, vol. 6, pp. 26231-26240, 2018.
- [49] L. Wang, S. Wang, Y. Xu, G. Zhou, and M. Liu, "A bi-population based estimation of distribution algorithm for the flexible job-shop scheduling problem," *Computers & Industrial Engineering*, vol. 62, no. 4, pp. 917-926, 2012.
- [50] R. H. Caldeira and A. Gnanavelbabu, "A. Solving the flexible job shop scheduling problem using an improved Jaya algorithm," *Computers & Industrial Engineering*, vol. 137, pp. 106064, 2019.
- [51] M. Alzaqebah, S. Jawarneh, M. Alwohaibi, M. K. Alsmadi, I. Almarashdeh, and R. M. A. Mohammad, "Hybrid brain storm optimization algorithm and late acceptance hill climbing to solve the flexible job-shop scheduling problem," *Journal of King Saud University-Computer and Information Sciences*, vol. 34, no. 6, pp. 2926-2937, 2022.
- [52] D. Yang, M. Wu, D. Li, Y. Xu, X. Zhou, and Z. Yang, "Dynamic opposite learning enhanced dragonfly algorithm for solving large-scale flexible job shop scheduling problem," *Knowledge-Based Systems*, vol. 238, pp. 107815, 2022.
- [53] C. Soto, B. Dorransoro, H. Fraire, L. Cruz-Reyes, C. Gomez-Santillan, and N. Rangel, "Solving the multi-objective flexible job shop scheduling problem with a novel parallel branch and bound algorithm," *Swarm and Evolutionary Computation*, vol. 53, pp. 100632, 2020.
- [54] S.-J. Wang, B.-H. Zhou, and L.-F. Xi, "A filtered-beam-search-based heuristic algorithm for flexible job-shop scheduling problem," *International Journal of Production Research*, vol. 46, no. 11, pp. 3027-3058, 2008.
- [55] M. A. F. Pérez and F. M. P. Raupp, "A Newton-based heuristic algorithm for multi-objective flexible job-shop scheduling problem," *Journal of Intelligent Manufacturing*, vol. 27, pp. 519-528, 2016.
- [56] J. Gao, L. Sun, and M. Gen, "A hybrid genetic and variable neighborhood descent algorithm for flexible job shop scheduling problems," *Computers & Operations Research*, vol. 35, no. 9, pp. 2892-2907, 2008.
- [57] X. Wang, L. Gao, C. Zhang, and X. Shao, "A multi-objective genetic algorithm based on immune and entropy principle for flexible job-shop scheduling problem," *Computers & Operations Research*, vol. 51, no. 5-8, pp. 757-767, 2010.
- [58] M. Bezoui, A. L. Olteanu, and M. Sevaux, "Integrating preferences within multiobjective flexible job shop scheduling," *European Journal of Operational Research*, vol. 305, no. 3, pp. 1079-1086, 2023.
- [59] A. Türkyılmaz, O. Senvar, and S. Bulkan, "A hybrid genetic algorithm based on a two-level hypervolume contribution measure selection strategy for bi-objective flexible job shop problem," *Computers & Operations Research*, vol. 141, no. 3, pp. 105694, 2022.
- [60] L. Yin, X. Li, L. Gao, C. Lu, and Z. Zhang, "A novel mathematical model and multi-objective method for the low-carbon flexible job shop scheduling problem," *Sustainable Computing: Informatics and Systems*, vol. 13, pp. 15-30, 2017.
- [61] H. Piroozfard, K. Y. Wong, and W. P. Wong, "Minimizing total carbon footprint and total late work criterion in flexible job shop scheduling by using an improved multi-objective genetic algorithm," *Resources, Conservation and Recycling*, vol. 128, pp. 267-283, 2018.
- [62] H. Wang, Z. Jiang, Y. Wang, H. Zhang, and Y. Wang, "A two-stage optimization method for energy-saving flexible job-shop scheduling based on energy dynamic characterization," *Journal of Cleaner Production*, vol. 188, pp. 575-588, 2018.
- [63] G. Zhang, X. Shao, P. Li, and L. Gao, "An effective hybrid particle swarm optimization algorithm for multi-objective flexible job-shop scheduling problem," *Computers & Industrial Engineering*, vol. 56, no. 4, pp. 1309-1318, 2009.
- [64] G. Moslehi and M. Mahnam, "A Pareto approach to multi-objective flexible job-shop scheduling problem using particle swarm optimization and local search," *Computers & Industrial Engineering*, vol. 129, no. 1, pp. 14-22, 2011.
- [65] X. Shao, W. Liu, Q. Liu, and C. Zhang, "Hybrid discrete particle swarm optimization for multi-objective flexible job-shop scheduling problem," *The International Journal of Advanced Manufacturing Technology*, vol. 67, pp. 2885-2901, 2013.
- [66] E. R. R. Kato, G. D. de A. Aranha, and R. H. Tsunaki, "A new approach to solve the flexible job shop problem based on a hybrid particle swarm optimization and Random-Restart Hill Climbing," *Computers & Industrial Engineering*, vol. 125, pp. 178-189, 2018.
- [67] M. R. Singh, M. Singh, S. S. Mahapatra, and N. Jagadev, "Particle swarm optimization algorithm embedded with maximum deviation theory for solving multi-objective flexible job shop scheduling problem," *The International Journal of Advanced Manufacturing Technology*, vol. 85, pp. 2353-2366, 2016.
- [68] J. Q. Li, Q. K. Pan, and K. Z. Gao, "Pareto-based discrete artificial bee colony algorithm for multi-objective flexible job shop scheduling problems," *The International Journal of Advanced Manufacturing Technology*, vol. 55, pp. 1159-1169, 2011.
- [69] L. Wang, G. Zhou, Y. Xu, and M. Liu, "An enhanced Pareto-based artificial bee colony algorithm for the multi-objective flexible job-shop scheduling," *The International Journal of Advanced Manufacturing Technology*, vol. 60, pp. 1111-1123, 2012.

- [70] J. Q. Li, Q. K. Pan, and M. F. Tasgetiren, "A discrete artificial bee colony algorithm for the multi-objective flexible job-shop scheduling problem with maintenance activities," *Applied Mathematical Modelling*, vol. 38, no. 3, pp. 1111-1132, 2014.
- [71] Y. Li, W. Huang, R. Wu, and K. Guo, "An improved artificial bee colony algorithm for solving multi-objective low-carbon flexible job shop scheduling problem," *Applied Soft Computing*, vol. 95, pp. 106544, 2020.
- [72] Y. Wang, J. Ge, S. Miao, T. Jiang, and X. Shen, "Application of hybrid artificial bee colony algorithm based on load balancing in aerospace composite material manufacturing," *Expert Systems with Applications*, vol. 215, pp. 119375, 2023.
- [73] L. Wang, "Multi-objective optimization based on decomposition for flexible job shop scheduling under time-of-use electricity prices," *Knowledge-Based Systems*, vol. 204, pp. 106177, 2020.
- [74] X. Wu and Y. Sun, "A green scheduling algorithm for flexible job shop with energy-saving measures," *Journal of cleaner production*, vol. 172, pp. 3249-3264, 2018.
- [75] Z. Wei, W. Liao, and L. Zhang, "Hybrid energy-efficient scheduling measures for flexible job-shop problem with variable machining speeds," *Expert Systems with Applications*, vol. 197, pp. 116785, 2022.
- [76] X. Sun, Y. Wang, H. Kan, Y. Shen, Q. Chen, and D. Wang, "Modified multi-crossover operator nsga-iii for solving low carbon flexible job shop scheduling problem," *Processes*, vol. 9, no. 1, pp. 62, 2020.
- [77] X. Gong, T. de Pesselier, L. Martens, and W. Joseph, "Energy-and labor-aware flexible job shop scheduling under dynamic electricity pricing: A many-objective optimization investigation," *Journal of cleaner production*, vol. 209, pp. 1078-1094, 2019.
- [78] J. Li, Q. Pan, and Y. C. Liang, "An effective hybrid tabu search algorithm for multi-objective flexible job-shop scheduling problems," *Computers & Industrial Engineering*, vol. 59, no. 4, pp. 647-662, 2010.
- [79] D. Lei, Y. Zheng, and X. Guo, "A shuffled frog-leaping algorithm for flexible job shop scheduling with the consideration of energy consumption," *International Journal of Production Research*, vol. 55, no. 11, pp. 3126-3140, 2017.
- [80] D. Lei, M. Li, and L. Wang, "A two-phase meta-heuristic for multiobjective flexible job shop scheduling problem with total energy consumption threshold," *IEEE Transactions on Cybernetics*, vol. 49, no. 3, pp. 1097-1109, 2018.
- [81] Q. Liu, M. Zhan, F. O. Chekem, X. Shao, B. Ying, and J. W. Sutherland, "A hybrid fruit fly algorithm for solving flexible job-shop scheduling to reduce manufacturing carbon footprint," *Journal of Cleaner Production*, vol. 168, pp. 668-678, 2017.
- [82] S. Karthikeyan, P. Asokan, S. Nickolas, and T. Page, "A hybrid discrete firefly algorithm for solving multi-objective flexible job shop scheduling problems," *International Journal of Bio-Inspired Computation*, vol. 7, no. 6, pp. 386-401, 2015.
- [83] Y. Yuan and H. Xu, "Multiobjective flexible job shop scheduling using memetic algorithms," *IEEE Transactions on Automation Science and Engineering*, vol. 12, no. 1, pp. 336-353, 2013.
- [84] N. Berti, S. Finco, O. Battaia, and X. Delorme, "Ageing workforce effects in dual-resource constrained job-shop scheduling," *International Journal of Production Economics*, vol. 237, pp. 108151, 2021.
- [85] X. Cao and Z. Yang, "An improved genetic algorithm for dual-resource constrained flexible job shop scheduling," *Proc. of Fourth International Conference on Intelligent Computation Technology and Automation*, pp. 42-45, 2011.
- [86] C. Peng, Y. Fang, P. Lou, and J. Yan, "Analysis of double-resource flexible job shop scheduling problem based on genetic algorithm," *Proc. of IEEE 15th International Conference on Networking, Sensing and Control*, pp. 1-6, 2018.
- [87] R. Wu, Y. Li, S. Guo, and W. Xu, "Solving the dual-resource constrained flexible job shop scheduling problem with learning effect by a hybrid genetic algorithm," *Advances in Mechanical Engineering*, vol. 10, no. 10, pp. 75-89, 2018.
- [88] S. Zhang, H. Du, S. Borucki, S. Jin, T. Hou, and Z. Li, "Dual resource constrained flexible job shop scheduling based on improved quantum genetic algorithm," *Machines*, vol. 9, no. 6, pp. 108, 2021.
- [89] D. Lei and X. Guo, "Variable neighbourhood search for dual-resource constrained flexible job shop scheduling," *International Journal of Production Research*, vol. 52, no. 9, pp. 2519-2529, 2014.
- [90] M. Yazdani, M. Zandieh, R. Tavakkoli-Moghaddam, and F. Jolai, "Two meta-heuristic algorithms for the dual-resource constrained flexible job-shop scheduling problem," *Scientia Iranica*, vol. 22, no. 3, pp. 1242-1257, 2015.
- [91] G. Gong, R. Chiong, Q. Deng, and X. Gong, "A hybrid artificial bee colony algorithm for flexible job shop scheduling with worker flexibility," *International Journal of Production Research*, vol. 58, no. 14, pp. 4406-4420, 2020.
- [92] X. Zheng and L. Wang, "A knowledge-guided fruit fly optimization algorithm for dual resource constrained flexible job-shop scheduling problem," *International Journal of Production Research*, vol. 54, no. 18, pp. 5554-5566, 2016.
- [93] L. Gao and Q. K. Pan, "A shuffled multi-swarm micro-migrating birds optimizer for a multi-resource-constrained flexible Job shop Scheduling Problem," *Information Sciences*, vol. 372, no. 1, pp. 655-676, 2016.
- [94] F. Madenoğlu and A. Baykasoğlu, "Dual-resource constrained flexible job shop scheduling problem with weighted superposition attraction algorithm," *Proc. of International Congress on Human-computer Interaction, Optimization and Robotic Applications*, pp. 167-170, 2019.
- [95] A. Gnanavelbabu, R. H. Caldeira, and T. Vaidyanathan, "A simulation-based modified backtracking search algorithm for multi-objective stochastic flexible job shop scheduling problem with worker flexibility," *Applied Soft Computing*, vol. 113, pp. 107960, 2021.

- [96] X. X. Liu, C. B. Liu, and Z. Tao, "Research on bi-objective scheduling of dual-Resource constrained flexible job shop," *Advanced Materials Research*, vol. 211, no. 1, pp. 1091-1095, 2011.
- [97] G. Gong, Q. Deng, X. Gong, W. Liu, and Q. Ren, "A new double flexible job-shop scheduling problem integrating processing time, green production, and human factor indicators," *Journal of Cleaner Production*, vol. 174, no. 1, pp. 560-576, 2017.
- [98] M. Yazdani, M. Zandieh, and R. Tavakkoli-Moghaddam, "Evolutionary algorithms for multi-objective dual-resource constrained flexible job-shop scheduling problem," *Opsearch*, vol. 56, no. 3, pp. 983-1006, 2019.
- [99] W. Tan, X. Yuan, J. Wang, and X. Zhang, "A fatigue-conscious dual resource constrained flexible job shop scheduling problem by enhanced NSGA-II: An application from casting workshop," *Computers & Industrial Engineering*, vol. 160, pp. 107557, 2021.
- [100] X. Gong, Q. Deng, G. Gong, W. Liu, and Q. Ren, "A memetic algorithm for multi-objective flexible job-shop problem with worker flexibility," *International Journal of Production Research*, vol. 56, no. 7, pp. 2506-2522, 2018.
- [101] H. Zhu, Q. Deng, L. Zhang, X. Hu, and W. Lin, "Low carbon flexible job shop scheduling problem considering worker learning using a memetic algorithm," *Optimization and Engineering*, vol. 21, pp. 1691-1716, 2020.
- [102] H. Lou, X. Wang, Z. Dong, and Y. Yang, "Memetic algorithm based on learning and decomposition for multiobjective flexible job shop scheduling considering human factors," *Swarm and Evolutionary Computation*, vol. 75, pp. 101204, 2022.
- [103] X. Li and Y. Liu, "Approach of solving dual resource constrained multi-objective flexible job shop scheduling problem based on MOEA/D," *International Journal of Online Engineering*, vol. 14, no. 7, pp. 75-89, 2021.
- [104] J. Shi, M. Chen, Y. Ma, and F. Qiao, "A new boredom-aware dual-resource constrained flexible job shop scheduling problem using a two-stage multi-objective particle swarm optimization algorithm," *Information Sciences*, vol. 643, pp. 119141, 2023.
- [105] L. Meng, Y. Ren, B. Zhang, J. Q. Li, H. Sang, and C. Zhang, "Milp modeling and optimization of energyefficient distributed flexible job shop scheduling problem," *IEEE Access*, vol. 8, pp. 191191-191203, 2020.
- [106] M. Ziaee, "A heuristic algorithm for the distributed and flexible job-shop scheduling problem," *The Journal of Supercomputing*, vol. 67, pp. 69-83, 2014.
- [107] F. T. Chan, S. H. Chung, and P. L. Y. Chan, "An adaptive genetic algorithm with dominated genes for distributed scheduling problems," *Expert Systems with Applications*, vol. 29, no. 2, pp. 364-371, 2005.
- [108] F. T. Chan, S. H. Chung, and P. L. Y. Chan, "Application of genetic algorithms with dominant genes in a distributed scheduling problem in flexible manufacturing systems," *International Journal of Production Research*, vol. 44, no. 3, pp. 523-543, 2006.
- [109] L. de Giovanni and F. Pezzella, "An improved genetic algorithm for the distributed and flexible job-shop scheduling problem," *European journal of operational research*, vol. 200, no. 2, pp. 395-408, 2010.
- [110] H. C. Chang and T. K. Liu, "Optimisation of distributed manufacturing flexible job shop scheduling by using hybrid genetic algorithms," *Journal of Intelligent Manufacturing*, vol. 28, no. 2, pp. 1973-1986, 2017.
- [111] P. H. Lu, M. C. Wu, H. Tan, Y. H. Peng, and C. F. Chen, "A genetic algorithm embedded with a concise chromosome representation for distributed and flexible job-shop scheduling problems," *Journal of Intelligent Manufacturing*, vol. 29, pp. 19-34, 2018.
- [112] W. Xu, Y. Hu, W. Luo, L. Wang, and R. Wu, "A multi-objective scheduling method for distributed and flexible job shop based on hybrid genetic algorithm and tabu search considering operation outsourcing and carbon emission," *Computers & Industrial Engineering*, vol. 157, pp. 107318, 2021.
- [113] Q. Luo, Q. Deng, G. Gong, L. Zhang, W. Han, and K. Li, "An efficient memetic algorithm for distributed flexible job shop scheduling problem with transfers," *Expert Systems with Applications*, vol. 160, pp. 113721, 2020.
- [114] Y. Sang and J. Tan, "Intelligent factory many-objective distributed flexible job shop collaborative scheduling method," *Computers & Industrial Engineering*, vol. 164, pp. 107884, 2022.
- [115] J. Q. Li, P. Duan, J. Cao, X. P. Lin, and Y. Y. Han, "A hybrid Pareto-based tabu search for the distributed flexible job shop scheduling problem with E/T criteria," *IEEE Access*, vol. 6, pp. 58883-58897, 2018.
- [116] Y. Du, J. Q. Li, C. Luo, and L. L. Meng, "A hybrid estimation of distribution algorithm for distributed flexible job shop scheduling with crane transportations," *Swarm and Evolutionary Computation*, vol. 62, pp. 100861, 2021.
- [117] M. Gholami and M. Zandieh, "Integrating simulation and genetic algorithm to schedule a dynamic flexible job shop," *Journal of Intelligent Manufacturing*, vol. 20, pp. 481-498, 2009.
- [118] N. Al-Hinai and T. Y. ElMekkawy, "Robust and stable flexible job shop scheduling with random machine breakdowns using a hybrid genetic algorithm," *International Journal of Production Economics*, vol. 132, no. 2, pp. 179-291, 2011.
- [119] W. He and D. Sun, "Scheduling flexible job shop problem subject to machine breakdown with route changing and right-shift strategies," *International Journal of Advanced Manufacturing Technology*, vol. 66, no. 1-4, pp. 501-514, 2013.
- [120] Y. M. Wang, H. L. Yin, and K. D. Qin, "A novel genetic algorithm for flexible job shop scheduling problems with machine disruptions," *International Journal of Advanced Manufacturing Technology*, vol. 68, no. 5-8, pp. 1317-1326, 2013.
- [121] Z. Y. Gu, L. F. Li, J. J. Zheng, and G. R. Liu, "A flexible job-shop rescheduling method by considering the machine equipment availability," *2016 28th Chinese Control and Decision Conference*, pp. 4898-4902, 2016.

- [122] J. Xiong, L. Xing, and Y. Chen, "Robust scheduling for multi-objective flexible job-shop problems with random machine breakdowns," *International Journal of Production Economics*, vol. 141, no. 1, pp. 112-126, 2013.
- [123] E. Ahmadi, M. Zandieh, M. Farrokh, and S. M. Emami, "A multi objective optimization approach for flexible job shop scheduling problem under random machine breakdown by evolutionary algorithms," *Computers & Operations Research*, vol. 73, pp. 56-66, 2016.
- [124] Y. Yang, M. Huang, Z. Yu Wang, and Q. Bing Zhu, "Robust scheduling based on extreme learning machine for bi-objective flexible job-shop problems with machine breakdowns," *Expert Systems with Applications*, vol. 158, p. 113545, 2020.
- [125] M. Nouiri, A. Bekrar, A. Jemai, D. Trentesaux, A. C. Ammari, and S. Niar, "Two stage particle swarm optimization to solve the flexible job shop predictive scheduling problem considering possible machine breakdowns," *Computers & Industrial Engineering*, vol. 112, pp. 595-606, 2017.
- [126] M. Nouiri, A. Bekrar, and D. Trentesaux, "Towards energy efficient scheduling and rescheduling for dynamic flexible job shop problem," *IFAC-PapersOnLine*, vol. 51, no. 11, pp. 1275-1280, 2018.
- [127] L. Wei, J. He, Z. Guo, and Z. Hu, "A multi-objective migrating birds optimization algorithm based on game theory for dynamic flexible job shop scheduling problem," *Expert Systems with Applications*, vol. 227, pp. 120268, 2023.
- [128] M. B. S. Sreekara Reddy, C. Ratnam, G. Rajyalakshmi, and V. K. Manupati, "An effective hybrid multi objective evolutionary algorithm for solving real time event in flexible job shop scheduling problem," *Measurement*, vol. 114, pp. 78-90, 2018.
- [129] R. Buddala and S. S. Mahapatra, "Two-stage teaching-learning-based optimization method for flexible job-shop scheduling under machine breakdown," *International Journal of Advanced Manufacturing Technology*, vol. 100, no. 5-8, pp. 1419-1432, 2019.
- [130] K. Z. Gao, P. N. Suganthan, Q. K. Pan, M. F. Tasgetiren, and A. Sadollah, "Artificial bee colony algorithm for scheduling and rescheduling fuzzy flexible job shop problem with new job insertion," *Knowledge-based Systems*, vol. 109, pp. 1-16, 2016.
- [131] K. Z. Gao, M. C. Zhou, and Y. X. Pan, "Jaya algorithm for rescheduling flexible job shop problem with machine recovery," *Proc. of IEEE International Conference on Systems, Man and Cybernetics*, pp. 3660-3664, 2019.
- [132] A. Teymourifar, G. Ozturk, Z. K. Ozturk, and O. Bahadir, "Extracting new dispatching rules for multi-objective dynamic flexible job shop scheduling with limited buffer spaces," *Cognitive Computation*, vol. 12, no. 1, pp. 195-205, 2020.
- [133] Z. Zakaria and S. Petrovic, "Genetic algorithms for match-up rescheduling of the flexible manufacturing systems," *Computers & Industrial Engineering*, vol. 62, no. 2, pp. 670-686, 2012.
- [134] L. Nie, L. Gao, P. Li, and X. Li, "A GEP-based reactive scheduling policies constructing approach for dynamic flexible job shop scheduling problem with job release dates," *Journal Of Intelligent Manufacturing*, vol. 24, no. 4, pp. 763-774, 2013.
- [135] K. Z. Gao, P. N. Suganthan, T. J. Chua, C. S. Chong, T. X. Cai, and Q. K. Pan, "A two-stage artificial bee colony algorithm scheduling flexible job-shop scheduling problem with new job insertion," *Expert Systems with Applications*, vol. 42, no. 21, pp. 7652-7663, 2015.
- [136] K. Li, Q. Deng, L. Zhang, Q. Fan, G. Gong, and S. Ding, "An effective MCTS-based algorithm for minimizing makespan in dynamic flexible job shop scheduling problem," *Computers & Industrial Engineering*, vol. 155, pp. 107211, 2021.
- [137] J. J. Palacios, M. A. González, C. R. Vela, I. González-Rodríguez, and J. Puente, "Genetic tabu search for the fuzzy flexible job shop problem," *Computers & Operations Research*, vol. 54, no. 10, pp. 74-89, 2015.
- [138] Y. Xu, L. Wang, S. Y. Wang, and M. Liu, "An effective teaching-learning-based optimization algorithm for the flexible job-shop scheduling problem with fuzzy processing time," *Neurocomputing*, vol. 148, pp. 260-268, 2015.
- [139] N. Xie and N. Chen, "Flexible job shop scheduling problem with interval grey processing time," *Applied Soft Computing*, vol. 70, pp. 513-524, 2018.
- [140] T. Jamrus, C. F. Chien, M. Gen, and K. Sethanan, "Hybrid particle swarm optimization combined with genetic operators for flexible job-shop scheduling under uncertain processing time for semiconductor manufacturing," *IEEE Transactions on Semiconductor Manufacturing*, vol. 31, no. 1, pp. 32-41, 2017.
- [141] R. Li, W. Gong, L. Wang, and C. Lu, "Two-stage knowledge-driven evolutionary algorithm for distributed green flexible job shop scheduling with type-2 fuzzy processing time," *Swarm and Evolutionary Computation*, vol. 74, pp. 101139, 2022.
- [142] R. Li, W. Gong, and C. Lu, "Self-adaptive multi-objective evolutionary algorithm for flexible job shop scheduling with fuzzy processing time," *Computers & Industrial Engineering*, vol. 168, pp. 108099, 2022.
- [143] Z. Pan, D. Lei, and L. Wang, "A bi-population evolutionary algorithm with feedback for energy-efficient fuzzy flexible job shop scheduling," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 52, no. 8, pp. 5295-5307, 2021.
- [144] P. Fattahi and A. Fallahi, "Dynamic scheduling in flexible job shop systems by considering simultaneously efficiency and stability," *Cirp Journal of Manufacturing Science & Technology*, vol. 2, no. 2, pp. 114-123, 2010.
- [145] A. Rajabinasab and S. Mansour, "Dynamic flexible job shop scheduling with alternative process plans: an agent-based approach," *International Journal of Advanced Manufacturing Technology*, vol. 54, no. 9-2, pp. 1091-1107, 2011.
- [146] L. Lin, M. Gen, Y. Liang, and K. Ohno, "A Hybrid EA for Reactive Flexible Job-shop Scheduling," *Procedia Computer Science*, vol. 12, pp. 110-115, 2012.

- [147] X.-N. Shen and X. Yao, "Mathematical modeling and multi-objective evolutionary algorithms applied to dynamic flexible job shop scheduling problems," *Information Sciences*, vol. 298, pp. 198-224, 2015.
- [148] T. Ning, M. Huang, X. Liang, and H. Jin, "A novel dynamic scheduling strategy for solving flexible job-shop problems," *Journal of Ambient Intelligence and Humanized Computing*, vol. 7, no. 5, pp. 721-729, 2016.
- [149] L. Zhang, X. Li, L. Gao, and G. Zhang, "Dynamic rescheduling in FMS that is simultaneously considering energy consumption and schedule efficiency," *International Journal of Advanced Manufacturing Technology*, vol. 87, no. 5-8, pp. 1387-1399, 2016.
- [150] X. X. Li, Z. Peng, B. G. Du, J. Guo, W. X. Xu, and K. Zhuang, "Hybrid artificial bee colony algorithm with a rescheduling strategy for solving flexible job shop scheduling problems," *Computers & Industrial Engineering*, vol. 113, pp. 10-26, 2017.
- [151] Y. Kou, C. Li, L. Li, Y. Tang, and X. Li, "Energy-efficient rescheduling for the flexible machining systems with random machine breakdown and urgent job arrival," *2019 IEEE International Conference on Systems, Man and Cybernetics*, pp. 409-414, 2019.
- [152] F. Psarommatis, A. Gharraei, and D. Kiritsis, "Identification of the critical reaction times for re-scheduling flexible job shops for different types of unexpected events," *Proceedia CIRP*, vol. 93, pp. 903-908, 2020.
- [153] Y. Li, Y. He, Y. Wang, F. Tao, and J. W. Sutherland, "An optimization method for energy-conscious production in flexible machining job shops with dynamic job arrivals and machine breakdowns," *Journal Of Cleaner Production*, vol. 254, pp. 120009, 2020.
- [154] A. Baykasoğlu, F. S. Madenoğlu, and A. Hamzaday, "Greedy randomized adaptive search for dynamic flexible job-shop scheduling," *Journal Of Manufacturing Systems*, vol. 56, pp. 425-451, 2020.
- [155] K. Z. Gao, F. J. Yang, J. Q. Li, H. Y. Sang, and J. P. Luo, "Improved Jaya algorithm for flexible job shop rescheduling problem," *IEEE Access*, 2020.
- [156] S. Luo, L. Zhang, and Y. Fan, "Dynamic multi-objective scheduling for flexible job shop by deep reinforcement learning," *Computers & Industrial Engineering*, vol. 159, pp. 107489, 2021.
- [157] P. Soofi, M. Yazdani, M. Amiri, and M. A. Adibi, "Robust Fuzzy-Stochastic Programming Model and Meta-Heuristic Algorithms for Dual-Resource Constrained Flexible Job-Shop Scheduling Problem Under Machine Breakdown," *IEEE Access*, vol. 9, pp. 155740-155762, 2021.
- [158] M. Ghaleb, S. Taghipour, and H. Zolfagharinia, "Real-time integrated production-scheduling and maintenance-planning in a flexible job shop with machine deterioration and condition-based maintenance," *J. Manuf. Syst.*, vol. 61, pp. 423-449, 2021.
- [159] H. Wang, Y. Jiang, H. Wang, and H. Luo, "An online optimization scheme of the dynamic flexible job shop scheduling problem for intelligent manufacturing," *2022 4th International Conference on Industrial Artificial Intelligence*, pp. 1-6, 2022.
- [160] J. Duan and J. Wang, "Robust scheduling for flexible machining job shop subject to machine breakdowns and new job arrivals considering system reusability and task recurrence," *Expert Systems with Applications*, vol. 203, pp. 117489, 2022.
- [161] Y. Lv, C. Li, Y. Tang, and Y. Kou, "Toward energy-efficient rescheduling decision mechanisms for flexible job shop with dynamic events and alternative process plans," *IEEE Transactions on Automation Science and Engineering*, vol. 19, no. 4, pp. 3259-3275, 2022.
- [162] Y. An, X. Chen, K. Gao, Y. L. and L. Zhang, "Multi-objective flexible job-shop rescheduling with new job insertion and machine preventive maintenance," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3101-3113, 2023.
- [163] Y. Gu, M. Chen, and L. Wang, "A self-learning discrete salp swarm algorithm based on deep reinforcement learning for dynamic job shop scheduling problem," *Applied Intelligence*, 2023.
- [164] G. Zhang, X. Lu, X. Liu, L. Zhang, S. Wei, and W. Zhang, "An effective two-stage algorithm based on convolutional neural network for the bi-objective flexible job shop scheduling problem with machine breakdown," *Expert Systems with Applications*, vol. 203, p. 117460, 2022.
- [165] H. Wang, J. Cheng, C. Liu, Y. Zhang, S. Hu, and L. Chen, "Multi-objective reinforcement learning framework for dynamic flexible job shop scheduling problem with uncertain events," *Applied Soft Computing*, vol. 131, pp. 109717, 2022.
- [166] S. Luo, L. Zhang, and Y. Fan, "Real-time scheduling for dynamic partial-no-wait multiobjective flexible job shop by deep reinforcement learning," *IEEE Transactions on Automation Science and Engineering*, vol. 19, no. 4, pp. 3020-3038, 2022.
- [167] A. Fadil, F. Chetouane, and Z. Binder, "Control of flexible job-shop: Disturbances and robustness," *IFAC Proceedings Volumes*, vol. 30, no. 19, pp. 13-18, 1997.
- [168] N. Zribi and P. Borne, "Hybrid genetic algorithm for the flexible job-shop problem under maintenance constraints," *Lecture Notes in Computer Science*, vol. 3612, pp. 259-268, 2005.
- [169] N. Zribi, A. El Kamel, and P. Borne, "Minimizing the makespan for the MPM job-shop with availability constraints," *International Journal of Production Economics*, vol. 112, no. 1, pp. 151-160, 2008.
- [170] E. Moradi, S. M. T. F. Ghomi, and M. Zandieh, "An efficient architecture for scheduling flexible job-shop with machine availability constraints," *International Journal of Advanced Manufacturing Technology*, vol. 51, no. 1-4, pp. 325-339, 2010.
- [171] J. Gao, M. Gen and L. Sun, "Scheduling jobs and maintenances in flexible job shop with a hybrid genetic algorithm," *Journal Of Intelligent Manufacturing*, vol. 17, no. 4, pp. 493-507, 2006.
- [172] M. Rajkumar, P. Asokan, and V. Vamsikrishna, "A GRASP algorithm for flexible job-shop scheduling with maintenance constraints," *International Journal of Production Research*, vol. 48, no. 22, pp. 6821-6836, 2010.

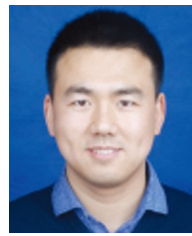
- [173] E. Moradi, S. M. T. F. Ghomi, and M. Zandieh, "Bi-objective optimization research on integrated fixed time interval preventive maintenance and production for scheduling flexible job-shop problem," *Expert Systems with Applications*, vol. 38, no. 6, pp. 7169-7178, 2011.
- [174] V. M. Dalfard and G. Mohammadi, "Two meta-heuristic algorithms for solving multi-objective flexible job-shop scheduling with parallel machine and maintenance constraints," *Computers & Mathematics With Applications*, vol. 64, no. 6, pp. 2111-2117, 2012.
- [175] J. Li and Q. Pan, "Chemical-reaction optimization for flexible job-shop scheduling problems with maintenance activity," *Applied Soft Computing*, vol. 12, no. 9, pp. 2896-2912, 2012.
- [176] J.-Q. Li, Q.-K. Pan, and M. F. Tasgetiren, "A discrete artificial bee colony algorithm for the multi-objective flexible job-shop scheduling problem with maintenance activities," *Applied Mathematical Modelling*, vol. 38, no. 3, pp. 1111-1132, 2014.
- [177] W. J. Song, C. Y. Zhang, W. W. Lin, and X. Y. Shao, "Flexible job-shop scheduling problem with maintenance activities considering energy consumption," *Applied Mechanics and Materials*, vol. 521, pp. 707-713, 2014.
- [178] M. Ziaee, "An efficient heuristic algorithm for flexible job shop scheduling with maintenance constraints," *Applied Mathematics and Sciences: An International Journal*, vol. 1, no. 1, pp. 19-31, 2014.
- [179] H. Mokhtari and A. Hasani, "An energy-efficient multi-objective optimization for flexible job-shop scheduling problem," *Computational Chemical Engineering*, vol. 104, pp. 339-352, 2017.
- [180] F. E. Khoukhi, J. Boukachour, and A. E. H. Alaoui, "The "dual-ants colony": A novel hybrid approach for the flexible job shop scheduling problem with preventive maintenance," *Computers & Industrial Engineering*, vol. 106, pp. 236-255, 2017.
- [181] S. H. A. Rahmati, A. Ahmadi, and K. Govindan, "A novel integrated condition-based maintenance and stochastic flexible job shop scheduling problem: Simulation-based optimization approach," *Annals Of Operations Research*, vol. 269, pp. 583-621, 2018.
- [182] N. Mirahmadi and S. Taghipour, "Energy-efficient optimization of flexible job shop scheduling and preventive maintenance," *Proc. of Annual Reliability and Maintainability Symposium*, 2019.
- [183] H. Wang, B. Sheng, Q. B. Lu, X. Y. Yin, F. Y. Zhao, X. C. Lu, R. P. Luo, and G. C. Fu, "A novel multi-objective optimization algorithm for the integrated scheduling of flexible job shops considering preventive maintenance activities and transportation processes," *Soft Computing*, vol. 25, no. 4, pp. 2863-2889, 2021.
- [184] Y. An, X. Chen, Y. Li, J. Zhang, and J. Jiang, "Flexible job-shop scheduling and heterogeneous repairman assignment with maintenance time window and employee timetable constraints," *Expert Systems with Applications*, vol. 186, pp. 115693, 2021.
- [185] M. M. Wocker, F. F. Ostermeier, T. Wanninger, R. Zwinkau, and J. Deuse, "Flexible job shop scheduling with preventive maintenance consideration," *Journal Of Intelligent Manufacturing*, 2023.
- [186] Y. Zheng, K. Mesghouni, and S. C. Dutilleul, "Condition based maintenance applied to reduce unavailability of machines in flexible job shop scheduling problem," *IFAC Proceedings Volumes*, vol. 46, no. 9, pp. 1405-1410, 2013.
- [187] Y. H. Zheng, L. Lian, and K. Mesghouni, "Comparative study of heuristics algorithms in solving flexible job shop scheduling problem with condition based maintenance," *Journal of Industrial Engineering and Management*, vol. 7, no. 2, pp. 518-531, 2014.
- [188] M. Zandieh, A. R. Khatami, and S. H. A. Rahmati, "Flexible job shop scheduling under condition-based maintenance: Improved version of imperialist competitive algorithm," *Applied Soft Computing*, vol. 58, pp. 449-464, 2017.
- [189] S. Wang and J. Yu, "An effective heuristic for flexible job-shop scheduling problem with maintenance activities," *Computers & Industrial Engineering*, vol. 59, no.3, pp. 436-447, 2010.
- [190] A. Baykasoğlu and F. S. Madenoğlu, "Greedy randomized adaptive search procedure for simultaneous scheduling of production and preventive maintenance activities in dynamic flexible job shops," *Soft Computing*, vol. 25, no. 23, pp. 14893-14932, 2021.
- [191] C. Chen, N. Y. Lu, B. Jiang, and Y. Xing, "Condition-based maintenance optimization for continuously monitored degrading systems under imperfect maintenance actions," *Journal of Systems Engineering and Electronics*, vol. 31, no. 4, pp. 841-851, 2020.
- [192] C. Chen, N. Y. Lu, B. Jiang, and C. Wang, "A risk-averse remaining useful life estimation for predictive maintenance," *Acta Automatica Sinica*, vol. 8, no. 2, pp. 412-422, 2022.
- [193] C. Wang, N. Y. Lu, Y. Cheng, and B. Jiang, "A data-driven aero-engine degradation prognostic strategy," *IEEE Transactions on Cybernetics*, vol. 51, no. 3, pp. 1531-1541, 2021.
- [194] Y. H. Xu, B. Jiang, and Y. Meng, "Adaptive dynamic programming approach for stackelberg game-based fault-tolerant control," *Proc. of 6th International Symposium on Autonomous Systems*, pp. 1-6, 2023.
- [195] Y. J. Ma, H. Ren, G. Tao, and B. Jiang, "Adaptive compensation for actuation sign faults of flexible spacecraft," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 57, no. 2, pp. 1288-1300, 2020.
- [196] Z. Zhao, "Sample-based dynamic event-triggered algorithm for optimization problem of multi-agent systems," *International Journal of Control, Automation, and Systems*, vol. 20, no. 8, pp. 2492-2502, 2022.
- [197] S. Liu, C. Duan, and W. Zhou, "Event-triggering control of networked control systems under random deception attacks: an adaptive triggering strategy with saturation constraint," *International Journal of Control, Automation, and Systems*, vol. 21, no. 9, pp. 2916-2926, 2023.

- [198] H. Yang, B. Jiang, M. Staroswiecki, and Y. Zhang, "Fault recoverability and fault tolerant control for a class of interconnected nonlinear systems," *Automatica*, vol. 54, pp. 49-55, 2015.
- [199] K. Zhang, B. Jiang, X. Yan, Z. H. Mao, and M. M. Polycarpou, "Fault-tolerant control for systems with unmatched actuator faults and disturbances," *IEEE Transactions on Automatic Control*, vol. 66, no. 4, pp. 1725-1732, 2020.
- [200] K. Zhang, B. Jiang, F. Y. Chen, and H. Yang, "Directed-graph-learning-based diagnosis of multiple faults for high speed train with switched dynamics," *IEEE Transactions on Cybernetics*, vol. 53, no. 3, pp. 1712-1724, 2021.
- [201] C. Cheng, X. Y. Sun, J. J. Shao, H. T. Chen, and C. Shang, "Just-in-time learning-aided nonlinear fault detection for traction systems of high-speed trains," *International Journal of Control, Automation, and Systems*, vol. 21, no. 9, pp. 2797-2809, 2023.
- [202] K. Hu, W. Sun, and Z. Cheng, "Improved adaptive compensation of variant fighter with multiple faults via extended observer," *International Journal of Control, Automation, and Systems*, vol. 21, no. 7, pp. 2277-2292, 2023.
- [203] W. Ren and S. Guo, "State and faults interval estimations for discrete-time linear systems," *International Journal of Control, Automation, and Systems*, vol. 21, no. 7, pp. 2303-2312, 2023.



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